

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics

In partial fulfillment for the requirements in Capstone 1 and
Research Subject

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CHAPTER 1

1.1 Background of the Study

In today's fast-paced world, businesses across all sectors are shifting to online platforms to keep up with the growing expectations of their consumers. The food service and enterprise baking industry is no exception to this trend. As tech-savvy individuals increasingly turn to digital applications for seamless, on-demand transactions, local micro-enterprises that continue to rely on manual, analog processes face significant challenges. Traditional, non-automated methods often result in critical operational inefficiencies, reduced market reach, and an inability to meet contemporary consumer demands. This shift in buyer behavior has turned digital convenience from a luxury into an absolute necessity for survival in the food retail industry. When small stores lack a modern web presence, they hiddenly lose customers to larger competitors who offer instant online ordering.

The underlying problem stems from how consumer data is collected and managed during initial touchpoints. In a typical digital commerce environment, data inputs are structured, validated, and processed instantaneously by backend application logic. Conversely, traditional store structures rely heavily on human

memory and fragmented paperwork to handle transactions.

This lack of automated validation means that business transactions are highly susceptible to data loss, tracking delays, and operational blind spots. For community-scale enterprises, continuing to operate within a purely analog framework creates a technical barrier that cuts them off from the modern digital marketplace, making it increasingly difficult to sustain consistent growth or build long-term customer loyalty.

C&N Bakery is a local bakery operating within a highly competitive local market, offering a wide selection of baked goods categorized into standard product groups, namely cakes, cupcakes, and traditional Filipino kakanin. In addition to mass retail pastries, the establishment specializes in bespoke, high-value custom cake orders and supports a pre-ordering framework designed for clients requiring specialized baked goods for celebrations, weddings, debuts, and milestones. Despite managing a diverse product line and catering to a growing clientele, C&N Bakery handles its entire business workflow through manual mechanisms, such as physical walk-ins, phone calls, and scattered social media messages, without any centralized digital infrastructure to sustain its operations. This lack of a single system means information is scattered

across different mediums, making daily tasks stressful and disorganized.

When a business does not use one main place to store its records, its information easily gets scattered across too many different places. At C&N Bakery, a single custom transaction often leaves a scattered data trail across multiple platforms: a Facebook Messenger text thread may contain the cake's design layout, a follow-up phone call might alter the delivery deadline, and a hand-written paper note on the counter might record the downpayment. Because these data points are completely unlinked, the administrative staff must manually piece together the customer's files before sending the order to the kitchen. This fragmented approach consumes massive amounts of time and makes day-to-day operations chaotic, turning simple routine tasks into exhausting manual procedures that drain the small business's daily energy.

The administrative overhead of managing custom cake orders is inherently more complex than standard point-of-sale retail transactions. Customizations demand precise recording of multiple structural variables, including layers, chiffon flavors, frosting variations, visual design styles, and custom dedication text, alongside strict coordination of pickup and delivery dates to preserve food freshness and structural

integrity. Relying on handwritten logbooks, paper receipts, and unorganized chat logs severely increases the risk of order leakage, a critical vulnerability where explicit structural design choices are forgotten or overlooked during preparation. Furthermore, human error in manual calendar logging leads to schedule overlaps, where production capacity is inadvertently exceeded for a specific date.

This operational vulnerability is deeply tied to the technical concept of data mutation. When a customer outlines their custom cake selections through a social media chat channel, that information exists as an unformatted text string. As that request moves from the front counter staff to the kitchen bakers, the information is manually rewritten onto paper order tags. Every time a human manually copies, interprets, or rewrites a data string, the higher the risk of transcription mistakes occurring. Essential design attributes—such as specific icing colors or custom text names—can easily be misread or completely forgotten during the transfer. This manual workflow creates a weak system loop where the original intent of the customer rarely matches the actual physical instructions given to the baking staff.

When the bakery relies on scattered chats and unorganized paper notes, it creates a high risk of transcription mistakes during

the order-taking process. These data entry errors directly lead to incorrect cake designs being sent to the kitchen. Once a cake is baked with the wrong design, flavor, or configuration, it triggers a chain reaction of operational failures across the business. This workflow breakdown results in wasted raw ingredients, disrupted baking queues, and unrecoverable financial losses for the bakery, while severely damaging long-term customer trust.

The financial damage caused by these data errors is made worse by the rigid nature of the baking process itself. Unlike other food service businesses that can quickly cook a replacement dish in a few minutes, a custom cake requires hours of uninterrupted baking, cooling, structural stabilization, and detailed frosting application. If a kitchen assistant discovers a configuration error late in the pipeline due to a manual data entry mistake, the entire product must often be discarded. This means the bakery loses both the expensive ingredients used and the valuable time slots in the oven schedule. This disruption delays every other order waiting in the queue, creating a stressful work environment and causing a backlog of delayed deliveries that ripples through the entire business day.

The operational difficulties are further worsened by the rapidly evolving commercial ecosystem in the Philippines, marked by the

widespread adoption of e-wallets and digital banking applications. Modern consumers expect transaction flows that offer instantaneous price validation, absolute cost transparency, interactive customer review modules, and structured frequently asked questions (FAQ) components, combined with the convenience of settling accounts immediately via QR codes or digital bank transfers. For C&N Bakery, maintaining a purely manual setup limits its ability to engage this expanding, digitally native market segment that prioritizes efficiency and remote ordering. When a customer has to wait hours for a manual message reply just to get a price quote, they often abandon the purchase entirely.

This bottleneck is a direct result of operating without an automated computational engine to handle basic pricing calculations. In a manual communication model, every customer inquiry regarding custom options forces a staff member to stop their physical kitchen tasks, sit down, and manually calculate costs line-by-line. This slow process creates a massive communication delay, leaving potential buyers waiting in digital queues for simple information. In the modern local economy, where consumers are accustomed to instant digital responses, these long waiting times cause high cart abandonment rates, as buyers quickly abandon the conversion funnel to buy from larger

automated competitors instead.

Beyond customer-facing constraints, the absence of a unified, web platform introduces severe security risks and data management obstacles. On the user side, clients lack an authenticated portal to manage personal accounts, execute secure password updates, or monitor the status of their orders. For administrative personnel, operating without a secure, role-based backend dashboard means that confidential business information—such as historical sales ledgers, client records, and sensitive operational statistics—remains vulnerable to data corruption, unbacked physical destruction, and unauthorized exposure, making data integrity a constant corporate concern. If a single paper logbook is misplaced or stained in the kitchen, the bakery completely loses its financial records and customer histories.

This total reliance on single-copy physical storage devices exposes the business to immense operational hazards. Paper logbooks kept inside an active kitchen are constantly exposed to water damage, grease stains, and accidental loss. From an information security perspective, storing a company's entire transaction history on unsecured paper slips leaves confidential client addresses and contact numbers open to unauthorized eyes. Furthermore, physical records offer zero data recovery options;

if a fire or major plumbing leak occurs in the shop, years of vital business history are permanently destroyed in minutes, leaving the management with no backup logs to reconstruct their operational data.

These combined systemic vulnerabilities generate persistent issues for both stakeholders. Customers endure a fragmented user experience due to the lack of an online catalog with dynamic filters to match specific budgets, or an intuitive design canvas to submit custom reference images. Concurrently, the production and administrative staff struggle to monitor simultaneous orders across various stages of readiness, which leads to scheduling errors and delivery delays. Without a visual tool to see how many cakes are currently in the oven, ready for frosting, or awaiting pickup, the small kitchen team easily becomes overwhelmed.

This internal visibility gap points to a clear need for synchronized multi-device tracking within the workspace. When independent teams (such as front desk clerks and back-office bakers) operate without a shared, real-time data view, they lose track of the overall business state. The baking staff has no clear way of knowing which orders are urgent, while the front counter team cannot accurately tell customers when their cakes will be ready for pickup. This total lack of internal connection

forces staff to constantly interrupt their workflows to check on order statuses verbally, which wastes valuable time, increases human stress, and leads to scheduling mistakes that frustrate waiting customers.

To resolve these operational bottlenecks and adapt to modern business trends, this study proposes the development of the Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics for C&N Bakery. This web-based system is built to provide an intuitive ordering framework for consumers while equipping bakery staff with a digital pipeline to monitor orders across precise production milestones. Ultimately, the system provides management with an automated scheduling engine to strictly enforce capacity thresholds and a visual sales analytics suite to capture monthly revenue performance and identify best-selling products. By transitioning to this unified setup, C&N Bakery can eliminate communication errors, secure its records, and grow its business in the local market.

1.2 Statement of the Problem

General problem statement

The primary operational challenge facing C&N Bakery is its heavy reliance on manual paperwork, physical logbooks, and unorganized social media chat messages. Because the bakery does not use a single, shared online system that updates everything at the same time, its data is completely split up. Crucial details like order counts, specific cake designs, and customer contact info are scattered across loose notes, paper kitchen calendars, and chaotic Facebook or Instagram inbox histories.

Relying on this unlinked, manual setup makes it incredibly easy to lose, misread, or accidentally destroy information. Without an automated web system to instantly capture and lock in order details the exact moment a customer checks out, the bakery lacks a single, reliable source of truth for its daily transactions. On a day-to-day basis, this broken process causes constant communication mix-ups during design consultations, costly cake assembly mistakes in the kitchen, expensive ingredient waste, and accidental calendar overbooking. Furthermore, keeping financial records and customer details in open paper notebooks leaves the bakery vulnerable to data privacy risks, accidental fluid spills, and permanent file loss. Ultimately, these issues prevent C&N Bakery from reaching tech-savvy online buyers, protecting its private records, and using automated analytics to grow its local business.

Specific statement of the Problem1.

1. Customer-facing interface and customization issues

- **Messy online menus and reply delays:**

C&N Bakery lacks a public digital catalog. When customers want to browse available pastries, check flavor variations, or ask about pricing, they are forced to send a private message or call the shop. Because communication relies entirely on manual replies from staff, the bakery experiences severe message backlogs during holidays and

weekends. Customers regularly wait hours for a basic quote, leading to frustration and lost sales. This chat-based setup acts as a bottleneck because a human worker can only reply to one person at a time, limiting sales speed by how fast an employee can type.

- **Pricing guesswork and math errors:**

Estimating the cost of a highly customized cake is difficult because prices change based on multiple selections. Currently, staff calculate quotes on the fly using rough mental math or loose price sheets while messaging clients. This lack of a structured system leads to inconsistent pricing, human errors, and a total lack of transparency for the buyer, who cannot see how changing options affects their bill. Without an automated pricing engine, two clients ordering the exact same cake profile can easily receive different quotes due to staff fatigue, which damages customer trust.

- **Misunderstood cake designs and lost order details:**

Custom cakes require tracking very exact details, like precise lettering spelling, chiffon flavors, and outer frosting styles. Currently, staff read through long chat screens, interpret messy handwriting, and copy these requirements by hand onto paper slips. This messy hand-off

creates a massive risk of missing or miscopying details.

When design requests are misread by the time the paper slip reaches the kitchen, it leads to incorrect cake assemblies, rejected orders, and a heavy waste of expensive baking ingredients.

- **Lack of public shop info and customer reviews :**

Modern online consumers rely on social proof and clear policies before sending digital payments for custom event cakes. C&N Bakery's manual setup provides no public platform for buyers to read past customer reviews, view product ratings, or access instant answers about delivery rules, allergen notices, or payment steps. This information vacuum causes heavy customer hesitation, as buyers are reluctant to send down-payments without seeing proof of past execution and clear shop guidelines upfront.

2. Administrative, scheduling, and kitchen operations failures

- **Kitchen overbooking and overlapping schedules:**

To maintain product quality, the kitchen has a strict capacity limit of three custom bakes per day. Currently, deadlines are tracked using a physical notebook calendar in the kitchen. Because different employees take orders simultaneously over the phone and social media without a shared view of the schedule, the bakery regularly

double-books dates. This overloads the kitchen team, leading to stressful rushes, dropped decorating quality, and late customer pickups.

- **Disorganized production progress and workspace tracking:**

Once an order is accepted, it must move sequentially through kitchen preparation stages. Currently, the kitchen staff has no shared visual tool to see where open orders stand in this timeline. Instead, progress is shared through verbal shouting or by sorting through loose paper notes. This forces kitchen staff to constantly stop their delicate decorating tasks to answer status questions from the front counter, creating a stressful work environment filled with constant interruptions.

- **Accidental early deliveries and confusion:**

Managing precise pickup and delivery dates is critical to keeping custom cakes fresh and intact. Under the current manual system, there are no safety checks to prevent scheduling mistakes. A staff member can easily misread a date on a paper slip and accidentally prepare or try to send out a delicate custom cake days before the customer actually expects it. This causes severe logistical confusion, communication mixed-ups.

Slow hand-offs and manual record keeping:

Once an order leaves the bakery for customer pickup or delivery, confirming that the sale was successfully finished requires manual follow-up calls or direct text messages. The bakery staff must then manually search for the original paper slip and write the payment details down into a physical ledger book. If a tired worker forgets to file a completed slip at the end of an exhausting shift, that entire sales record effectively vanishes from the day's bookkeeping, creating financial blind spots.

3. Data privacy, analytics, and financial recording

- **Unsecured business records and lack of user privacy:**

Operating without a secure database leaves C&N Bakery open to severe data privacy risks. Customers have no private, password-protected profile page where they can safely save their contact details, view past orders, or check billing receipts. On the business side, confidential financial logs, past sales records, and customer contact lists are kept on unbacked paper notebooks or shared personal mobile phones, leaving sensitive business data completely exposed to physical destruction, spills, total loss, or unauthorized.

- **Slow payment tracking and accounting errors:**

Managing down-payments, cash balances, and Cash-on-Delivery (COD) collections using manual notebooks is incredibly time-consuming and prone to math mistakes. Because proof-of-payment screenshots from digital wallets (like GCash) are kept loose in mobile phone photo galleries, matching an uploaded payment slip to a specific paper order requires tedious manual cross-checking. This disconnected process leads to missing payment logs, unbalanced financial books, and frustrating money discrepancies at the end of the day.

- **Absence of clear business insights and profit metrics:**

Under the current manual system, the bakery owner cannot easily check if the business is growing or calculate profit margins without spending hours doing manual math at the end of every month. The owner has no automated way to view overall performance data, such as completed-versus-cancelled order ratios, busiest ordering hours, or precise sales breakdowns for each product category. To see how the bakery is doing, the owner must manually sort through and add up hundreds of individual paper order slips line by line. This data blindness leaves management unable to track changing customer preferences,

predict busy seasonal rushes, or identify which specific pricing tiers (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, or ~~P~~501+) make the most money, forcing them to rely on pure guesswork instead of real facts.

1.3 Objectives of the Study

1.3.1 General Objective

The main goal of this study is to design, build, launch, and test a secure web-based Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics for C&N Bakery. This system is created to get rid of the daily headaches caused by manual paperwork, scattered chat messages on social media, and messy paper calendars in the kitchen. By replacing these outdated habits with a single, synchronized website, the project aims to make tracking orders easy, give customers clear and upfront pricing, keep the kitchen from taking on too many orders at once, and provide the owner with simple sales charts for better business planning.

This project is driven by the urgent need to move a local home bakery into the digital age. Managing custom cake orders is a complicated process because staff must collect precise design details and match them perfectly against a tight baking schedule. By introducing this platform, the study aims to gather customer profiles, cake specifications, reference photos, and delivery details into a single, secure online database. This new setup will cut down on exhausting paperwork, stop staff from misreading handwritten design instructions, and protect the bakery from losing money on wasted ingredients. Ultimately, this

study provides C&N Bakery with a stable, easy-to-use platform that handles daily order traffic, keeps personal customer data safe, and helps a small local business thrive in today's digital market.

1.3.2 Specific Objectives

To successfully reach the main goal of this project, the study focuses on meeting these specific practical and design objectives:

- **Building the Interactive Product Menu and Custom Cake Pricing**

To create an easy-to-use online store menu that groups items into clear categories (Cakes, Cupcakes, and Kakanin) and includes simple budget filters (~~P0~~–~~P199~~, ~~P200~~–~~P500~~, and ~~P501~~+) so customers can browse effortlessly. Along with this, to build an automatic pricing tool inside the custom cake builder page that instantly recalculates the total cost as the customer clicks through four simple options: the number of layers, the cake flavor, the frosting type, and the design style.

- **Setting Up Photo Uploads and Trust-Building Information**

To add a secure file upload section where customers can attach their own reference cake design photos and type out exact text inscriptions during checkout. At the same time,

to build a public customer review section and an FAQ page so new visitors can quickly find clear answers about delivery guidelines, payment policies, and allergen notices without waiting hours for a manual text reply.

- **Developing the Smart Scheduling Calendar and multi-stage Kitchen Tracker**

To build an automated booking calendar on the admin side that uses clear color-coding to highlight busy days and automatically locks out a date once it hits a strict maximum limit of three (3) active custom orders. This calendar will connect directly to an order management board that divides open orders into clean production steps (**Pending, Confirmed, Baking, Ready, Completed,** and **Cancelled**) so the baking team can update a cake's progress with a quick click.

- **Creating Shipping Safeguards and Instant Order Archive Buttons**

To build helpful warning pop-up alerts that block staff from accidentally marking a cake as delivered or picked up before its actual scheduled date. This objective also includes creating a final confirmation button that automatically pushes finished orders into a permanent history log, ensuring that the system's sales records update the exact millisecond an order is closed out.

- **Securing User Logins, Account Roles, and the Payment Ledger**

To set up a secure login system that completely separates public customer accounts from private administrative control panels using role-based access rules. This objective also involves building an organized payment ledger table for the manager that features instant search bars and status dropdowns, making it incredibly easy to cross-check uploaded GCash receipt screenshots, track down-payments, and monitor Cash-on-Delivery (COD) balances safely.

- **The Owner's Sales Dashboard and Visual Trend Charts**

To build a real-time statistical dashboard for the bakery owner that displays data through a Monthly Revenue Chart and a Top Products Chart. This turns raw database logs into instant, readable graphs, allowing the owner to track monthly profit trends and pinpoint the shop's best-selling items immediately without doing manual math.

1.4 Scope and Delimitation

This study focuses on creating, coding, and launching a secure online ordering and scheduling website made specifically for C&N Bakery. The website is split into two main sections that share the same database:

The first interface is a Customer Portal, which is built to give buyers complete independence when looking through menus, estimating custom cake prices, and sending in order details. The second interface is a secure Administrative Dashboard, which is built to give the bakery owner and kitchen staff full visual control over daily kitchen limits, multi-step baking progress, menu item switches, and automated sales data charts.

The core structure of this platform brings together different business operations into a single online workspace. In the customer area, the platform features an organized product catalog with simple price navigation filters, a step-by-step custom cake creator that automatically updates prices on the screen, a secure button to upload reference images, a booking calendar with strict date capacity limits, a private account tracker page, and public info pages for common questions and reviews.

On the administrative side, the system connects directly to

these client inputs, displaying quick business health calculation cards, an organized order summary table with a built-in search index, a split-screen detail view that shows instruction sheets alongside design photos, a 5-step kitchen tracking bar, a quick toggle switch to handle out-of-stock items, a strict 3-order daily limit, and automated sales reporting charts.

1. System platform and accessibility

The system is built as a fully responsive website rather than a downloadable application. This development approach uses flexible grid structures and scalable layouts, ensuring that the entire interface adjusts dynamically to fit a wide range of hardware screens. The website delivers smooth layouts, text formatting, and full button functionality on desktop computers, tablets, and mobile phones.

Both customers and administrative staff can interact with the system using standard web browsers, including Google Chrome, Firefox, Apple Safari, and Brave .

2. Target users of the system

The system's operational features are explicitly built to match the day-to-day needs of three distinct target user groups:

- **Bakery consumers (Public and Registered users):**

This group includes regular visitors and registered members who use the platform to browse retail offerings, get instant custom cake price quotes, place pre-orders, and check their cake's production status without having to call or text the shop.

- **Production Staff (Kitchen Personnel):**

Workers and cake decorators who interact with the admin panel to view upcoming cake requirements, look at design reference images, and advance order status markers to let the front counter know what is happening in the kitchen.

- **Bakery owner (System Administrator):**

The business owner holds complete system privileges. They use the control panel to add or remove menu items, adjust base prices for ingredients, manually change locked calendar dates, and look at automated monthly financial graphs to see how the business is growing.

3. Customer portal functional scope

The user-facing part of the platform covers all the steps required to guide a customer from discovering the menu to completing a secure checkout and tracking their order:

- **Easy-to-Use menu and budget filters:**

This section displays a clean, organized product menu that separates items into clear groups: Cakes, Cupcakes, and Kakanin. To save customers from endless scrolling, it includes a simple text search bar and quick budget filters (₱0-₱199, ₱200-₱500, and ₱501+) so they can find options that match their wallet instantly.

- **4-Step custom cake creator:**

This feature gives customers a simple, step-by-step page to design their cake by choosing four basic details: how many layers they want, the cake flavor, the frosting type, and the design theme. As they click through these choices, the system automatically updates the final price on the screen in real time so there is zero guesswork.

- **Media asset upload system:**

To make sure the kitchen staff perfectly understands a customer's ideas, this module provides a secure file upload area that accepts high-resolution design photos (JPEG or PNG). It also includes a text field for custom cake inscriptions and dedications, preventing miscommunications, misspelled text, or forgotten design requests.

- **Smart fulfillment checkout form:**

When ready to buy, the customer uses an interactive calendar grid that displays available booking dates. The form collects the customer's contact details, including a direct link to their Facebook profile (optional), and lets them pick their fulfillment preference (Pick-up or Delivery) and payment type (Cash-on-Delivery, GCash, or Bank Transfer).

- **Private customer portal and order tracker:**

Logged-in customers get access to a private profile page featuring a live progress bar. This tool shows them exactly where their order stands (Pending, Unpaid, or Cash on Delivery) in real time, so they never have to call or message the shop just to ask for an update. It also includes a simple settings menu where users can safely update their account passwords.

- **Public information and review pages:**

To help new users order with confidence, this module includes an interactive customer review ledger where verified buyers can post ratings, along with an easy-to-read Frequently Asked Questions (FAQ) page covering pickup details, warnings, and payment validation steps.

4. Administrative dashboard functional scope

The back-end dashboard gives the bakery management complete control over business records, kitchen capacity rules, and sales analytics:

- **Owner's quick summary dashboard:**

Positioned right at the top of the admin page, this panel displays easy-to-read summary cards that track how the business is doing. It shows live, automated counts for Total Sales Made, Total Orders Processed, and the current number of Active Pending Orders so the owner can check the shop's status.

- **Organized order tracking table:**

This workspace features a clean, tab-sorted order table paired with a fast search bar to manage all open orders easily. Staff can view neatly arranged rows showing customer names, specific pastry choices, promised delivery times, total bill amounts, and payment verification tags without any struggles.

- **Split-screen order detail view :**

When an admin clicks on an active custom order, the screen opens into a clean, split layout. The left side displays

the customer's uploaded reference image, while the right side shows the structural cake options and custom text directions. This dual-view setup lets the cake decorator look at the design photo and the specific instructions at the exact same time without switching tabs.

- **5-Stage kitchen progress bar:**

This progress manager allows kitchen staff to move active orders through five sequential milestones: *Preparing*, *Baking*, *Packing*, and *Delivery*. It also features simple payment status checkboxes (Paid vs. Unpaid) to help management keep track of the bakery's cash flow.

- **Quick Menu On/Off Switches:**

This section features a central menu setting page where staff can click a simple on/off toggle switch next to any pastry. If the kitchen runs out of ingredients for a specific item, switching it "Off" instantly hides it from the public shop page, stopping customers from ordering treats that are currently sold out.

- **3-Order Daily Limit System:**

This smart booking tool keeps track of the kitchen's daily workload. The calendar automatically counts upcoming

bookings and locks out specific dates for customers the moment that day hits a strict limit of three (3) active custom cake orders, completely protecting the baking team from overbooking.

- **Owner's Sales Charts and Graphs:**

This secure reporting page takes raw transaction data and automatically turns it into two clear charts: a Monthly Revenue Chart and a Top Products Chart. This lets the owner spot profit trends and best-selling pastries instantly without doing any manual math.

Delimitation of the Study

- **No Native Mobile Apps:**

The system is built and optimized strictly for web browsers. This project will not create, build, or launch separate mobile applications for download on the Apple App Store or Google Play Store.

- **No Direct Payment Connections (APIs):**

While the checkout page shows the bakery's GCash QR code and bank details, the website does not connect directly to banks or e-wallet companies. The system cannot automatically check if a customer's money went through; verifying payments is a manual job where staff must look at

the customer's uploaded receipt photo and double-check it against the order.

- **No Third-Party Delivery Integration:**

The platform operates independently of outside delivery apps. It does not connect to delivery booking systems. All physical deliveries and driver bookings must be handled manually by the bakery outside of the website.

- **No Raw Ingredient Inventory Tracking:**

The system does not automatically calculate baking ingredients by weights or volumes (like kilograms of flour or liters of milk remaining). Instead, it relies on the staff manually using the on/off switches to hide menu items when ingredients run low.

- **No Employee Payroll or HR Tools:**

The software is strictly meant for managing orders and sales. It does not include human resource features like staff attendance logging, daily time cards, payroll computations, or shift scheduling for the employees.

- **No Supplier or Vendor Management:**

The website does not include features for logging

ingredient supplier contact info, generating bulk purchase orders, or tracking fluctuating wholesale ingredient costs from vendors.

- **No Offline Ordering Mode:**

It requires a live internet connection to load the database, meaning it does not support offline data saving or ordering when disconnected from the internet.

1.5 Significance of the Study

Building the Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics brings great value to everyone involved. As the food and bakery industry moves online, traditional neighborhood bakeshops must update their tools to keep up with competitors. This study shows exactly how a local bakery can escape the stress and limitations of manual paperwork. By replacing easily lost handwritten logbooks, sticky notes, and scattered social media chats with a single, synchronized website, this project creates a smooth ordering process. This new system ensures honest pricing for customers, protects the kitchen from accidental overbooking, and keeps important business records safe from getting damaged or lost. By improving these daily tasks, this study provides clear benefits to a wide group of people:

End Users (Bakery Customers and Guests)

Traditionally, everyday buyers trying to order custom cakes from local shops face frustrating customer service issues, such as scattered chat histories, long response delays on social media, and hidden pricing. This forces customers to explain complex cake designs through endless, confusing text messages. The launch of this online platform completely removes these communication roadblocks, directly helping customers through

these easy-to-use tools:

- **Honest and Instant Pricing**

Customers get complete price transparency through an automatic pricing calculator. This tool updates the total cost on the screen the exact moment a user clicks through different cake layers, flavors, frostings, or design styles, removing any guesswork.

- **Easy Photo Sharing and Accurate Designs**

Customers can use a secure upload button to attach their cake reference photos directly to their order form. By saving these photos alongside their custom text inscriptions, the system prevents design misunderstandings and forgotten details when the kitchen builds the cake.

- **Independent Order Tracking**

Users get their own private, password-protected account page with a live progress timeline. This shows the exact status of their order (**Pending, Unpaid, or Cash on Delivery**), removing the need to message the shop or wait hours for a staff reply.

- **Menu Browsing**

The website includes a clean online catalog with convenient price filters (**P0-P199, P200-P500, and P501+**) paired with a

helpful FAQ page. This is accessible 24/7 on any phone or computer, letting users shop whenever they want.

Administrative Control Personnel (Bakery Staff and Decorators)

Kitchen staff and cake decorators in a manual bakery setup work under a lot of daily stress. They are constantly forced to track upcoming deadlines using easily ruined paper notebooks, loose sticky notes, and calendar books. This disorganized setup regularly causes accidental overbooking, late orders, and expensive ingredient waste. This online platform introduces clean organization to the kitchen workspace:

- **No More Kitchen Overbooking**

Staff can track the total daily workload through a central, color-coded admin calendar. The system automatically enforces a strict limit of three (3) active custom orders per day, blocking further customer bookings on that date once the limit is met.

- **Zero Assembly and Design Mistakes:**

Cake decorators can view exact design instructions and customer-uploaded photos side-by-side using a split-screen detail window. This displays the customer's reference image on the left side of the screen and their custom text choices on the right, letting staff check the photo and

directions at the same time without flipping through loose papers.

- **Clear Daily Labor Priorities:**

Kitchen staff can easily organize daily tasks using a 4-stage kitchen progress timeline that monitors orders through **Preparing, Baking, Packing,** and **Delivery** states. By clicking simple action buttons beneath the timeline, staff can update order statuses instantly.

- **Instant Control of Menu Items**

The dashboard provides a settings page with simple on/off switches next to every pastry. If the kitchen runs out of ingredients for a specific product, clicking the switch instantly hides that item from the public website, preventing customers from ordering sold-out treats.

Executive Management (Bakery Owner)

For the business owner, running a shop without a central website makes daily financial tracking, profit auditing, and long-term planning incredibly difficult. Calculating monthly statistics manually using paper ledgers and calculators increases the risk of human math errors and leaves historical sales trends trapped on loose pieces of paper. This secure web platform protects the owner's daily business operations:

- **Automated Financial Records**

The owner gets quick access to an organized transaction history table equipped with fast search bars and status dropdown menus, making the tracking of down-payments, cash records, and GCash receipts seamless and error-free.

- **Clear Visual Sales Charts**

The website includes a central dashboard that automatically reads database histories and turns them into two easy-to-read graphs: a vertical Monthly Revenue Chart and a horizontal Top Products Chart, allowing the owner to spot profit trends and best-selling items instantly without doing manual math at the end of the month.

- **Data Protection and Privacy**

The platform uses secure login gates and role-based permissions to ensure that sensitive financial information, customer contact details, and internal business metrics are visible *only* to the owner, keeping business records safe from unauthorized eyes.

- **Smarter Business Decisions**

By providing accurate, unalterable digital records of customer purchases and payment preferences, the platform helps the owner make smarter choices regarding ingredient

budgets, wholesale shopping, and seasonal menu updates.

The Organization (C&N Bakery as a Business Entity)

Moving from outdated paper habits to an automated online platform helps professionalize C&N Bakery's local brand. It actively protects the shop's commercial reputation from the negative fallout of missed pickup windows, misspelled cake text, or forgotten order details. As a business storefront, the bakery benefits from several key improvements:

- **Prevention of Revenue and Ingredient Loss**

The system minimizes expensive kitchen waste and unpaid custom orders by enforcing strict checkout rules, collecting clear customer contact links, and tracking down-payment status steps before baking begins.

- **Stronger Local Competitiveness**

The platform grants a competitive edge, allowing a small local bakery to stand out in a highly competitive market by offering modern digital options—like instant online customization, digital receipts, and public review pages—that tech-savvy buyers expect.

- **Long-Term Growth Potential**

By streamlining internal kitchen workflows and automating

order tracking, the platform allows the bakery to handle significantly larger order volumes during peak holiday seasons without needing to hire extra administrative staff just to reply to messages.

Future Systems Developers

For software engineers and web developers building e-commerce websites for specialized food markets, generic online stores lack the strict booking limits required for custom cake baking. This study serves as a highly practical reference for building web applications that require strict, data-driven calendar restrictions. This research assists future developers in several distinct technical ways:

- **Detailed Product Customizer Reference**

The documentation serves as a clear architectural blueprint for coding interactive customizer screens that link specific, chosen product options directly to real-time pricing scripts.

- **Demonstration of Calendar Limit Patterns**

The project offers a real-world programming example for setting up calendar constraint gates that automatically check database rows and lock specific dates from further user inputs once a capacity threshold is met.

- **Future Academic Researchers**

This study helps researchers who are studying how small businesses in the Philippines shift from manual paper logs to simple websites. It provides real business workflows that can be compared to other small retail shops. It also gives a clear example of how to study a local shop's problems and design a web application specifically for made-to-order businesses. Lastly, it suggests easy next steps for future student projects, like adding automated SMS text alerts, direct online payment links, or rider delivery app integrations.

- **The Local Food Service Industry**

The success of this platform proves that helpful websites are affordable and easy to use for neighborhood bakeshops and small restaurants. It shows that family-owned bakeries do not need a massive budget to use an online catalog, a secure database, and an automated booking calendar. This project also helps raise local service standards by showing customers the benefits of clear pricing, data privacy, and real-time order tracking, which encourages other nearby food vendors to update their shops too.

- **BSIT Students**

For IT students, this documentation serves as a direct

guide for capstone projects and system defenses. It shows how to break down manual, messy business problems into clean and workable system features. It also explains how to turn actual storefront steps into clear database statuses and order timelines. Lastly, it gives a practical example of good user interface design by showing how to create clean pages, simple buttons, and secure login screens for an actual working website.

1.6 Definition of Terms

1. Web-Based System

- **General Definition:** A system that runs on a web browser and can be accessed online without needing to install any software on the user's device.
- **What it Does in Our System:** This refers to the online platform built for C&N Bakery that allows customers, staff, and the owner to access the menu, place orders, track baking progress, and view sales charts using any smartphone, tablet, or laptop without downloading an app.

2. User Authentication and Login

- **General Definition:** A security feature that checks a user's identity and account type before letting them into specific parts of a website.
- **What it Does in Our System:** This acts as a digital checkpoint. It reads whether an account belongs to a customer, a staff member, or the owner, ensuring customers only see the public shop page while locking down sensitive financial records and kitchen dashboards from unauthorized visitors.

3. Custom Cake Builder

- **General Definition:** An interactive webpage tool that lets

users select and customize different product options before checking out.

- **What it Does in Our System:** This is the page where customers design their custom cakes by picking the number of layers, flavors, frosting types, and design themes, while typing out text inscriptions and uploading their reference photos.

4. Automated Pricing Calculator

- **General Definition:** A script that instantly computes and updates the total price on a screen as different product options are selected.
- **What it Does in Our System:** It runs inside the Custom Cake Builder page. The moment a customer changes a cake layer size or flavor, the system updates the price quote on their screen in real time, so they do not have to wait for a manual text reply from the shop.

5. Online Menu Catalog

- **General Definition:** A digital storefront page where business products are organized into clear categories with search and filter tools.
- **What it Does in Our System:** This is the public browsing page for C&N Bakery. It organizes items into clean sections

(Cakes, Cupcakes, and Kakanin) and includes budget filters (P0-P199, P200-P500, and P501+) so customers can find items that match their budget easily.

6. Private Customer Dashboard and Live Tracker

- **General Definition:** A secure, private profile page where a logged-in user can manage their account details and view the status of their transactions.
- **What it Does in Our System:** This is the private portal for customers. Once logged in, they can update their password and view a live progress bar showing whether their active order is **Pending, Unpaid, or Cash on Delivery.**

7. Admin Summary Dashboard

- **General Definition:** A high-level overview panel at the top of a management page that displays automatically calculated business statistics and totals.
- **What it Does in Our System:** Positioned at the very top of the owner's admin page, this panel displays easy-to-read cards showing live counts for Total Sales, Total Orders Processed, and Active Pending Orders at a single glance.

8. Sales Analytics Charts

- **General Definition:** A reporting module that takes raw transaction numbers from a database and turns them into

visual charts and graphs.

- **What it Does in Our System:** This tool reads past order histories and automatically generates two clean visual graphs—a vertical Monthly Revenue Chart and a horizontal Top Products Chart—letting the owner spot sales trends without doing manual math.

9. Tabbed Order Management Table

- **General Definition:** A data table organized by tabs and search bars to help staff view, sort, and track active business workflows.
- **What it Does in Our System:** This is the master workspace on the admin page. Staff can click different tabs to view orders based on their status, using a search bar to instantly find customer names, pastry choices, delivery times, and payment tags.

10. Side-by-Side Order Detail View

- **General Definition:** A split-screen window layout that displays two related sets of information next to each other for easy reference.
- **What it Does in Our System:** When a staff member clicks a custom order, the screen splits. The left side displays the customer's uploaded reference photo, and the right side

shows the text instructions, allowing the cake decorator to look at both at the same time.

11. Menu Availability Switch

- **General Definition:** An administrative on/off button used to quickly hide or show a specific item on a public website.
- **What it Does in Our System:** This is a simple on/off switch next to every pastry in the admin menu. If the kitchen runs out of ingredients for an item, staff can click the switch to instantly hide it from the public shop page so customers cannot place accidental orders for sold-out treats.

12. 3-Order Daily Limit System

- **General Definition:** A scheduling system tool that monitors daily reservation limits and locks out calendar dates once the maximum threshold is met.
- **What it Does in Our System:** This tool manages the kitchen's daily workload. It automatically counts upcoming custom bookings and locks out specific dates on the customer's checkout calendar the moment that day hits a strict limit of three (3) active custom cake orders, preventing the team from being overbooked.

13. Kitchen Progress Timeline

- **General Definition:** A step-by-step visual tracker that maps

out a baking or production process into a time-ordered sequence of steps updated by the user.

- **What it Does in Our System:** This is the four-stage progress line (**Preparing, Baking, Packing, and Delivery**) used by the bakers. By clicking simple buttons beneath the timeline, staff can update the status of an order, which instantly refreshes the customer's tracking view.

CHAPTER 2

2.1 Foreign Studies

Shafiei, Khonsari, and Mousavi (2022) confirm that transitioning from fragmented, traditional hardware-bound server architectures to a dedicated serverless computing framework significantly improves system management, resource efficiency, and software deployment speed. Their research emphasizes that moving away from monolithic, legacy setups to a centralized cloud architecture allows for the automated handling of computational infrastructure, which prevents the heavy operational maintenance, constant framework upgrades, and manual operating system patching often associated with running physical, on-site hardware. The system paradigm was examined across multi-dimensional modules, exploring how serverless spaces utilize Function-as-a-Service (FaaS) and Backend-as-a-Service (BaaS) ecosystems to cleanly isolate application business logic inside cloud-level micro-containers, allowing development teams to focus entirely on running code smoothly without the low-level administrative burden of configuration tuning or security sandboxing.

The study's technical evaluation phase involved an in-depth analysis of dynamic scalability modeling across parallel and distributed workloads, focusing heavily on execution latency

bottlenecks like "cold starts"—which occur when an idle function is triggered after a long resting period—and evaluating modern optimization techniques, such as keeping container images warm or running lightweight pre-warmed background micro-runtimes to reduce initial response delays.

Furthermore, the researchers dive deep into the specific operational realities of containerization, documenting how traditional server environments inherently trap application code inside fixed operating systems that require ongoing human oversight, while serverless frameworks shift machine management, security isolate sandboxing, physical hardware upkeep, and underlying database maintenance entirely to the service provider. This structural decoupling removes the low-level administrative burden from developers, reducing architectural friction and minimizing software vulnerabilities caused by mismatched system dependencies or delayed server patches. The authors detail the dynamic scalability mechanisms of cloud environments, explaining that serverless platforms provide high speed and inherent dynamic scalability to efficiently handle highly dynamic, parallel, and distributed workloads. When an application experiences a massive wave of incoming traffic, the cloud framework executes parallel computing tasks by instantly spinning up isolated micro-container runtime instances, which

successfully avoids the CPU exhaustion, memory leakage, and request queue congestion that typically freeze traditional servers during traffic surges.

Later, when traffic volumes drop and the system returns to a resting state, the serverless framework optimizes financial resource utilization by automatically spinning down the idle micro-containers, preventing unnecessary server runtimes.

Additionally, the researchers provide a detailed evaluation of the pricing issue and financial structures found in serverless deployment spaces, highlighting the value of fine-grained cost models compared to traditional hosting paradigms. Traditional hosting web backends lock small operations into fixed, recurring leasing fees based on projected peak usage limits, meaning organizations pay full price for computational infrastructure even during slow hours when their applications are idle.

Serverless monetization models, by contrast, operate on a highly granular pay-as-you-go scale that measures performance by calculating execution time down to the exact millisecond alongside the specific memory size used per query. This ensures that when an application is sitting idle, operational expenses drop to zero.

The study concludes that the serverless platform provides clear architectural advantages for application developers looking to

eliminate local infrastructure management, making it an incredibly robust, secure, and cost-efficient environment for deploying modern, highly interactive web systems.

This study is highly relevant to the development of the C&N Bakery platform because it provides the technical reasons for using a serverless system. By using Google Firebase as our Backend-as-a-Service (BaaS) system, the web application does not need a local physical server setup. This completely removes the hard work of building and fixing server hardware, which perfectly aligns with the authors' points about reducing operational complexity. This setup is highly practical for a small, local business like C&N Bakery because they do not have an on-site IT department or a dedicated database team to look after heavy hardware. By running everything through Firebase's cloud services, the system stays online 24/7 without the physical risks and security problems that come with old-fashioned server setups.

The authors' findings on how serverless systems handle big spikes in web traffic directly solve a major problem for C&N Bakery. The bakery experiences massive changes in order volumes throughout the year. They have normal, quiet traffic on regular weekdays, but they get hit with an overwhelming number of custom cake orders and image uploads during major holidays like

Christmas, New Year, Valentine's Day, and Mother's Day. A normal web host might crash if too many customers try to customize cakes and check out at the exact same time. By using the serverless auto-scaling feature proven in this study, our platform can handle these busy holiday rushes smoothly by instantly pulling in more cloud power to keep the pages loading fast and ensure no customer's order or payment gets lost.

Lastly, the pay-as-you-go cost model discussed in the study makes the C&N Bakery platform highly affordable and sustainable. Traditional web hosting forces a business to pay an expensive, flat monthly fee even if nobody is using the website. Firebase, on the other hand, only charges the bakery for the exact amount of database reads, writes, and file transfers that happen when people actually use the application. This means during quiet business weeks, the system costs almost nothing to maintain.

Furthermore, the study's section on event-driven functions explains how our automatic emails work: the moment a baker updates an order's progress on the dashboard, a background cloud function instantly triggers and sends an email receipt to the customer's inbox without slowing down the rest of the website.

Sulova (2023) confirms that transitioning from fragmented, legacy digital storefronts to a comprehensively integrated and technologically advanced e-commerce framework significantly improves operational performance, data consistency, and organizational scalability for small and medium-sized business networks. The research emphasizes that a centralized, multi-layered system integration methodology prevents the information loss, manual administrative bottlenecks, and data disjointedness often associated with running disconnected operational tools and loose transactional systems. The conceptual framework was examined across precise core developmental directions, exploring the application of modern software tools, the enhancement of the foundational database base, and the dynamic possibilities of total architectural connectivity. The study's technical evaluation phase utilized a highly rigorous research methodology, leveraging advanced computer natural language processing and empirical text-mining analysis via platforms like RapidMiner Studio to systematically process a vast repository of scientific literature and isolate the core performance drivers of electronic retail performance. The findings showed that e-commerce optimization must be executed across three explicit levels of connectivity: application integration, data-level integration, and external

partner system integration. At the application integration level, the researcher documents how modern web stores must directly bundle primary user screens with auxiliary automated components, such as automated messaging paths, custom recommendation loops, and customer communication workflows. At the data-level integration stage, the study argues extensively for the deployment of a unified database infrastructure that actively merges unstructured transactional data streams, successfully eliminating isolated informational silos and ensuring that any data-driven event occurring on the frontend user portal maps directly to backend tracking tools in real time. To achieve true systemic unity, the paper highlights that database entries should not exist as single, unlinked rows but should actively communicate across every administrative view. This structural integration allows operational fields—such as order state transitions, real-time balance metrics, and customer contact parameters—to be updated instantly across the entire application ecosystem when a target field is toggled. By binding frontend inputs to structured backend cloud parameters, the architecture ensures absolute data alignment and complete software resilience against system failures.

Furthermore, the author deeply investigates how system integration can completely change how businesses handle their

records. The study breaks down how manually moving data from one software to another introduces heavy human error and tires out business staff. Sulova outlines a system layout where transaction histories are parsed instantly by automated cloud codes. This background process works day and night, evaluating item categories and checking totals without requiring a person to type anything into an Excel sheet. By designing automated pathways that constantly clean up and calculate incoming order metrics, the application can host active visual dashboards that show business charts instantly. This gives managers immediate access to live updates, allowing them to spot changing consumer habits and tweak their selling plans right away. Ultimately, the researcher proved that pairing client-facing shopping sites with unified backend management workspaces drastically cuts down the time spent on administrative tasks, removes clerical mistakes, increases tracking clarity, and hands small businesses the professional tools they need to grow and succeed in highly competitive digital markets.

This study is highly relevant to our project because it provides the practical blueprint we need to build the system design for the C&N Bakery platform. Our application directly solves the problem of scattered data by merging the customer menu catalog, the 4-choice cake customizer, the smart scheduling calendar, and

the manager's dashboard into a single, unified website. This matches Sulova's core concept of removing disconnected software silos. By putting all these features into one codebase, we can eliminate the bakery's old reliance on physical paper books and messy social media chat logs. This clean digital environment makes managing orders simple and effectively cuts the bakery's historical manual data entry mistakes down to zero.

The study's focus on connecting frontend user screens directly to live backend databases supports our technical programming choices. When a customer uses the C&N Bakery storefront, the system uses JavaScript to instantly package the custom design parameters from the cake order form into a clean JSON data structure. This data is pushed immediately into a serverless NoSQL database. Because the database is fully integrated with the manager's screen, the order details pop up on the kitchen dashboard right away without requiring the page to be manually reloaded. This real-time synchronization prevents lost orders and ensures that the kitchen staff can see exactly what options the customer chose when the second checkout is completed.

Additionally, the study's section on automated data integration supports the development of our Revenue Management Canvas. Instead of forcing the bakery owner to spend hours reading through manual pen-and-paper journals at the end of the month,

our platform uses background calculation functions to automatically read the transaction history. The system changes raw database entries into easy-to-read charts, showing vertical monthly sales bars and horizontal graphs of top-selling products. This gives the owner clear visual tools to track daily growth and see which cake flavors or sizes generate the highest profit margins, completely removing the risk of manual math errors. Furthermore, this structural dashboard update helps the owner stay blind-spot free, allowing management to track busy customer trends, prepare the right amount of ingredients, and plan for long-term expansion using real business facts instead of pure guesswork.

Pereira and do Prado (2011) confirm that transitioning from fragmented, rigid software engineering methods to a dedicated agile development framework significantly improves requirements engineering, user workflow alignment, and design flexibility for web applications. Their research emphasizes that an iterative development process prevents project delays, communication gaps, and software failures often associated with large upfront manuals and static specification sheets. The agile process was examined across three distinct disciplines, exploring how development teams can systematically refine user stories by pairing them with a structured Navigation Model and physical Story Cards to guide application coding. The study's technical evaluation phase utilized a highly rigorous methodology, tracking requirement lifecycles across collaborative web engineering paradigms, Behavior Driven Development, Test Driven Development, Scrum, and eXtreme Programming to achieve high real-world system adoption rates. The findings showed that breaking software requirements down using the 3 C's principle (Card, Conversation, and Confirmation) and the INVEST criteria accelerates construction by focusing strictly on features that bring direct value to the end-user.

At the planning level, the researchers document how a spatial Navigation Model allows developers to map out user paths and

track exactly how information flows between separate user interface panels before a single line of backend logic is written, uncovering hidden interaction gaps or broken pathways early in the development lifecycle. At the technical validation stage, the study argues extensively that using Story Cards enforces clear software boundaries by establishing concrete data fields and validation rules directly based on how workers perform their daily tasks.

Furthermore, the authors deeply investigate how running continuous feedback loops and user conversations improves the design of interactive user interfaces, allowing developers to optimize data fields and functional modules to match changing client workflows. By implementing this highly participative approach, the researchers concluded that the system not only dropped administrative friction and lowered software development costs but also increased overall system trust and customer satisfaction by delivering a digital product that smoothly handles real-world business routines.

This study is highly relevant to our project because it provides a strong justification for using an Agile development lifecycle to build the C&N Bakery platform. Because a small bakery's operational setups and menu options can easily change, using an iterative sprint model allowed us to adjust the website's

features as the bakery owner and staff tested early versions. This collaborative approach prevented project delays and kept us from building features the business did not actually need. To keep development organized, we divided the project into three distinct sprint windows: Sprint 1 delivered the public storefront with its price-bracket sorting canvas (P0-P199, P200-P500, and P501+); Sprint 2 focused on the core manager workspace, the 4-stage kitchen order tracker, and the split-screen image layout; and Sprint 3 tied everything together by integrating the automated analytics and email systems.

The study's focus on gathering requirements based on real-world business constraints supports our field data collection strategy. Instead of making generic assumptions about how an e-commerce store should work, our team spent time inside C&N Bakery's kitchen to map out their exact manual routines and physical capacity limits. This hands-on analysis gave us the precise rules needed to program smart software constraints. For example, we used these findings to build the Smart Scheduling Matrix Calendar, which automatically blocks out specific pickup dates on the customer checkout screen the moment a day hits its maximum capacity limit of three (3) custom cake orders.

Finally, the validation rules discussed in the paper guided our User Acceptance Testing (UAT) phase. By testing the live website

alongside the bakery owner, bakers, counter clerks, and ten regular retail customers, we gathered direct, practical feedback on how the system performs under real working conditions. This step helped us fine-tune the Split-Screen Layout Blueprint inside the order details window. By listening to the staff's input, we ensured that the cake decorators can clearly view custom design notes and customer reference photos side-by-side on a single screen, preventing eye strain and eliminating any confusion during busy kitchen production shifts.

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2.2 Local Studies

Abellana and Alonzo (2025) confirm that transitioning from fragmented, manual transaction methods to a dedicated, automated digital framework significantly improves operational efficiency, revenue protection, and business growth rates for micro, small, and medium-sized bakery enterprises in the Philippines. Their research emphasizes that relying on traditional, open-counter selling and inconsistent pen-and-paper record-keeping routines leads to data loss, chaotic ingredient tracking, and a total inability to monitor real business growth patterns. The system boundaries of local baking operations were examined using a highly rigorous descriptive-quantitative research methodology that systematically analyzed the operational realities and distribution pathways of 113 small independent bakeries across regional economic environments. The findings showed that entrepreneurial human capital—specifically the traits of business agility, operational adaptability, and technological complementarity—exerts a highly statistically significant and positive impact on an enterprise's actual survival rate, while the modernization of distribution channels serves as an essential structural driver for scaling business operations.

At the administrative level, the researchers document how external disruptions and physical mobility limitations severely damage traditional, face-to-face baking models, forcing business

owners to actively throw away manual logbooks that frequently get lost, broken, or misread. The study notes that under traditional methods, bakery managers remain completely unable to plan daily baking batches or calculate order balances effectively, especially when dealing with fluctuating market prices for raw ingredients like flour, sugar, and yeast. To resolve these structural operational roadblocks, the research places special focus on a complete transition toward digital transformation readiness, explicitly proving that combining automated management tools with digital storefront channels drops administrative friction and completely eliminates human tracking errors.

Furthermore, the authors deeply explore the broader socioeconomic difficulties that local bakeries face when managing day-to-day work tasks during unexpected economic shifts. The study expands on the fact that small bread and pastry shops operate on thin profit margins, meaning that even minor mistakes—like overbaking slow-selling goods or forgetting a customer's advanced deposit—can quickly ruin a shop's weekly cash flow. Abellana and Alonzo detail how introducing localized technology gives managers a direct look into their actual performance trends, showing them exactly which products move fast and which ones sit on shelves too long. By removing human

guesswork and setting up a secure, unfragmented digital pipeline, local food producers can smoothly adjust their daily baking schedules to match real market demands. Ultimately, the researchers concluded that moving away from old-fashioned open-counter setups not only protects small businesses from losing track of their sales but also builds long-term operational resilience, allowing local bakeries to professionalize their operations and safeguard their earnings in a highly competitive local market ecosystem.

This local study is highly relevant to our project because it provides direct, real-world proof of the exact business struggles currently facing C&N Bakery. Just like the 113 bakeries surveyed in Mindanao, C&N Bakery operates as a small-scale pastry enterprise that is held back by old-fashioned paper logs and face-to-face ordering habits. The authors' proof that inconsistent paper record-keeping stops a bakery owner from tracking real business growth perfectly justifies why our project is throwing away C&N Bakery's manual logbooks and replacing them with a centralized, automated web dashboard.

The study's findings on the need for "modernizing distribution channels" directly support our design choices for the public storefront. C&N Bakery cannot grow if it only relies on walk-in customers and physical menus. Our web platform addresses this

limitation by giving the bakery a complete online storefront. This allows customers to browse products, check pricing tiers via our sorting canvas (~~P0-P199~~, ~~P200-P500~~, and ~~P501+~~), and send order details from their mobile devices or computers. This system creates a clean, digital distribution loop that removes the physical barriers highlighted in the study.

Finally, our choice to link this storefront to a serverless Google Firebase backend answers the authors' call for "automated systems and digital transformation." Instead of the bakery staff spending hours listing sales numbers by hand, our database system works quietly in the background. It records transactions automatically and instantly feeds them into our visual data tools, such as the vertical monthly sales charts. This integration gives C&N Bakery an efficient way to manage order pipelines, calculate its true daily revenues, and avoid financial tracking errors without needing complex technical training.

Ballena, Caday, Iman, and Rolle (2022) confirm that transitioning from fragmented, unstructured social media messaging setups to a dedicated, structured web-based ordering framework significantly improves operational consistency, minimizes financial losses from communication errors, and reduces administrative burnout for micro-scale baking entrepreneurs in the Philippines. Their research emphasizes that relying on informal, loose chat logs for custom pastry transactions leads to chaotic information-gathering loops, missed structural details, and expensive production mistakes that ruin small business cash flows. The operational boundaries and lifecycles of local baking shops were examined using a basic interpretive qualitative research design paired with purposeful sampling, allowing the investigators to run systematic thematic extraction via open, axial, and selective coding to map the actual working realities of small-scale cake and bread sellers. The findings showed that while commercial social media platforms provide a free marketing channel to reach local consumer groups, they introduce severe data processing bottlenecks over time that actively stall long-term financial growth and strain small business systems.

At the production level, the researchers document how independent cake makers often work as one-person operations

where a single individual must handle raw ingredient sanitation, dough mixing, oven runtime monitoring, intricate icing decoration, customer messaging, and courier rider coordination simultaneously. The study notes that under traditional, chat-based methods, bakers experience extreme physical and mental stress during holiday peaks and weekend celebration rushes because they are forced to spend hours answering disjointed chat notifications instead of focusing on maintaining pastry quality and food safety tracking. To resolve these structural operational roadblocks, the research places special focus on moving away from unstructured messaging environments, explicitly proving that standardizing consumer transaction paths through a centralized, data-locked system dramatically lowers administrative friction and prevents the human clerical errors that frequently lead to costly product waste. By implementing this fixed digital pipeline, the researchers concluded that the system not only protected customer brand loyalty but also increased overall kitchen safety and business sustainability, delivering an efficient digital environment where incoming transactions instantly convert into organized, error-free kitchen instructions the moment an online order is cleared.

Furthermore, the authors deeply evaluate how these manual communication gaps directly translate into negative public

reviews and direct business loss when custom cakes are baked with the wrong design parameters, incorrect tier sizes, mismatched frosting colors, or mistaken delivery times. Because social media apps do not enforce required input fields, buyers frequently forget to state critical assembly parameters—such as explicit flavor bases, filling types, custom name inscriptions, or target pickup dates—forcing sellers to navigate confusing, time-consuming back-and-forth text exchanges that stretch over several days. In many tracked cases, local bakers were forced to absorb the heavy materials cost of remaking an entire multi-tiered cake from scratch or offer full cash refunds due to simple misread text strings inside a cluttered message inbox. Ultimately, the researchers proved that establishing a specialized web portal with strict operational validation limits serves as an essential tool to safeguard earnings, streamline delivery scheduling, and rescue local pastry entrepreneurs from the heavy operational anxieties commonly found in the fast-moving online food market ecosystem.

This local study is highly accurate and directly fits our project because it targets the exact operational headaches that C&N Bakery deals with every day. Currently, C&N Bakery takes custom cake requests through informal social media chat logs, which completely matches the exact messy scenario described by

the authors. By proving that unstructured chat messages cause communication blunders and expensive cake mistakes, this study gives our BSIT Capstone team a rock-solid reason to build a dedicated ordering platform for the bakery.

Our system directly solves the study's findings on "communication gaps" by taking the human guesswork out of the ordering process. Instead of letting a customer type a random, confusing message in a chat box, our website forces them to use a structured, 4-choice custom cake builder. When a user chooses their options, the frontend code instantly translates those clicks into a clean JSON data package. This ensures that essential details—like the cake flavor, size, and icing theme—are permanently locked into the system before the order is even sent to the kitchen. This completely eliminates the chaotic back-and-forth texting loops highlighted by Ballena et al.

Additionally, the study emphasizes how much local cake bakers struggle with stress when managing preparation timelines and order delivery slots. This insight perfectly justifies why we engineered the Smart Scheduling Matrix Calendar and the 4-stage kitchen stepper pipeline for C&N Bakery. By programming the system calendar to automatically block out specific dates the exact second they hit a maximum limit of three (3) custom orders, our application addresses the timeline anxieties

discussed in the study. It protects the bakers from accidental overbooking and structures the kitchen's daily workflow, giving the business the exact digital stability the authors recommend.

Omowole and Sari (2025) confirm that transitioning from fragmented, manual record-keeping methods to a dedicated, digitally integrated customer relationship management framework significantly improves customer retention, pricing accuracy, and long-term business survival within the local cake industry. Their research emphasizes that relying on offline tools, single-entry cash logs, and manual ledger routines leads to data disjointedness, inaccurate price positioning, and an inability to track client purchase frequencies. The operational boundaries of regional cake enterprises were examined using a quantitative and strategic management approach that gathered empirical data from both consumers (comprising a 75% respondent share) and independent bakery owners (comprising a 25% respondent share). By applying rigorous correlation metrics to this dataset, the researchers successfully evaluated how system-driven operational features influence consumer buying habits. The findings showed that long-term customer satisfaction and brand loyalty are heavily driven by fair, structured pricing models paired with active digital engagement strategies.

At the planning and market level, the researchers document how the modern local pastry market is dominated by young, digitally active individuals who prioritize affordability, customization flexibility, and transparent online brand spaces. The study

notes that under traditional, un-digitized methods, clients face immense friction when trying to check menu pricing against their immediate budgets, which frequently leads to hidden transactional charges and reduced customer trust. To resolve these structural operational roadblocks, the research places special focus on a complete transition toward digital transformation readiness, explicitly proving that organizing product offerings into clear, structured online categories reduces administrative friction and boosts client trust. By implementing a clear digital queue, the researchers concluded that the system not only improved backend tracking visibility but also handed small-scale baking operations a highly efficient way to manage order pipelines, calculate true daily revenues, and enhance overall market competitiveness in an evolving local market landscape.

Furthermore, the authors deeply explore the broader operational complexities that neighborhood bakeries face due to the multi-layered tracking requirements of custom cake assembly. Unlike standard pre-packaged retail pastries that follow simple, predictable inventory timelines, custom cakes require dynamic ingredient measurement, highly flexible production schedules, and strict coordination of pickup windows to guarantee freshness. Omowole and Sari detail how independent shops often

struggle with hidden financial losses during peak holiday rushes because their offline tracking routines prevent them from measuring immediate ingredient cost changes, calculating historical customer retention rates, or pinpointing which creative design themes yield the best financial returns. Without structured backend processing software, bakery managers are forced to guess their retail pricing, resulting in severe material waste and human accounting errors. Ultimately, the researchers proved that establishing a specialized web portal with strict database validation limits serves as an essential tool to safeguard earnings, streamline delivery scheduling, and rescue local pastry entrepreneurs from the heavy operational anxieties commonly found in the fast-moving online food market ecosystem.

This local study is highly accurate and directly fits our project because it targets the exact operational headaches that C&N Bakery deals with every day. Currently, C&N Bakery takes custom cake requests through informal social media chat logs and paper notebooks, which completely matches the messy scenario described by the authors. By proving that manual tracking stops a bakery owner from calculating true business growth, this study gives our BSIT Capstone team a rock-solid reason to build a dedicated ordering platform for the bakery.

Our system directly solves the study's findings on consumer budget anxieties by taking the human guesswork out of the ordering process. Instead of letting customers feel confused about hidden charges, our website forces the public menu to be organized around clear price-bracket sorting choices, specifically dividing items into easy filters like (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, and ~~P~~501+). This design choice ensures that local cake buyers can instantly filter products by their budget, matching local market expectations and building total brand trust right from the very first click.

Additionally, the study emphasizes how much local cake shops struggle with hidden losses because they lack automated data tools to view their business metrics or track custom product categories. Our Capstone platform addresses this exact issue by running background calculation functions on our cloud NoSQL database. The system automatically transforms raw sales entries into immediate visual tools—like vertical monthly sales charts and horizontal top-selling product graphs. This integration gives C&N Bakery an efficient way to manage order pipelines, calculate its true daily revenues, and avoid financial tracking errors without needing complex technical training.

Finally, this automated layout helps the bakery owner stay completely blind-spot free. Instead of wasting hours reading

through manual pen-and-paper journals at the end of the month, the owner can use the real-time manager dashboard to instantly see which cake designs and pricing tiers drive their highest profit margins. This allows the business to prepare the right amount of baking ingredients, adjust menu items on the fly, and plan for long-term expansion using real business facts instead of pure guesswork.

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2.3 RELATED LITERATURE

Modern information systems built for the local retail sector increasingly employ scalable, mobile-friendly frameworks to modernize manual operations and address severe delivery roadblocks. **Mindula et al. (2022)**, in their study titled **"Cake-Hut: A Novel E-Commerce Platform for Booking and Purchasing Cakes Online" (ISSN): 2250-0758** , highlight that shifting small businesses away from physical point-of-sale setups allows community shops to handle dynamic market demands without spending money on expensive, on-premise computing hardware. This technical design utilizes lightweight, document-oriented storage engines that execute queries on demand. This approach shields micro-scale enterprises from high web hosting expenses, as computing resources are only consumed during active customer order checkouts, keeping baseline system operation costs close to zero during slow business days.

The authors expand significantly on the structural design of document-oriented NoSQL database environments within specialized food markets. Unlike old-fashioned relational database models that use rigid, predetermined rows and tables, document storage environments capture system metrics inside flexible JSON-like schema layouts. Mindula et al. track how this technical design allows developers to store inconsistent user options, varying

pastry weights, custom design requests, and dynamic price variables together within a single document file. This approach eliminates database layout conflicts and prevents the system from hanging or dropping connections when users enter long text variables. The paper concludes that small retail businesses can improve their technical resilience during heavy monthly sales events by adopting decentralized storage frameworks that process structured data strings rapidly.

To further investigate structural database optimization for local micro-scale bakeshops, Mindula et al. dedicate an extensive discussion to the operational failures caused by running traditional SQL database management systems on low-tier web networks. The text documents that small retail food businesses frequently deal with unpredictable customer request strings, especially during peak holiday seasons, major institutional celebrations, and family events. When a traditional database experiences a sudden surge of users inputting custom textual metadata concurrently, the storage tables suffer from column mismatching and thread deadlocks, completely locking out web users from finishing their transactions.

The researchers highlight that by moving to non-relational document data stores, the backend structure frees itself from

strict database dependencies. Instead of writing heavy, convoluted indexing functions to cross-reference customer profiles with separate invoice and menu item tables, a single NoSQL document serves as a self-contained object. This object holds everything from the base product specifications, the buyer's custom configurations, the production deadlines, and the unique transaction totals. Mindula et al. prove that this encapsulation dramatically limits memory consumption and read-write input/output bottleneck cycles. This ensures that a localized, low-overhead digital system continues to run smoothly and processes data securely even when operating on unstable, consumer-grade broadband networks common across provincial areas.

This research study is relevant to our current project because it matches the exact technical challenges we are solving for C&N Bakery. In the Philippines, small local bakeries usually avoid setting up custom websites because hiring a technical team to maintain massive, traditional database servers is far too expensive for their limited budgets. Mindula's research proves that a business can get rid of these high monthly server maintenance fees completely by using an on-demand system. This heavily supports our team's choice to build the C&N Bakery platform using Firebase. Since our system only activates and

processes resources when a customer is online clicking through our catalog, the bakery does not have to worry about heavy monthly software fees during slow business days, keeping their overhead costs completely manageable.

Furthermore, the author's detailed breakdown of flexible, document-based files is the perfect justification for our primary feature: the 4-choice custom cake builder page. When customers order custom cakes, every single transaction is completely unique because users want different sizes, specialized frosting themes, custom text inscriptions, and specific flavor variations. Traditional databases easily crash or drop connections when users type unexpected text or combine strange options because those systems expect every single order to look exactly the same. Following the findings of Mindula's study, our platform saves every single custom cake order as a flexible data package inside Firebase. When a user creates a cake design on our storefront, Firebase instantly bundles those unique textual requirements into a self-contained object. This keeps our ordering page lightning-fast, avoids database layout conflicts, and ensures that the custom ordering system remains entirely crash-proof for everyday users.

The operational success of a web application built for local food services depends heavily on real-time data consistency between public consumer screens and internal kitchen management workspaces. **Kumar et al. (2026), in their study "Smart Canteen Application for Digital Order Management and Real-Time Status Tracking" (ISSN): 2582-3930** , emphasize that immediate state synchronization functions as a primary bridge between customer checkout actions and internal kitchen teams, offering total data abstraction and operational consistency across different frontend views. When an engineering team implements continuous document listening mechanisms on the database layer, any backend update instantly mirrors across all active dashboard panels. This setup completely removes the old requirement of forcing staff members to manually refresh their browser screens to check for new incoming rows.

The authors provide a deep technical analysis of how real-time synchronization pipelines reduce human communication blunders within fast-paced local bakery environments. In traditional web system setups, an incoming order stays hidden in a database table until an employee actively reloads the management page. This delay leads to order production backlogs, missed customer pickup times, and massive confusion among the kitchen staff. Kumar et al.'s literature proves that building live web sockets

or continuous data channels allows systems to transform raw database changes into instant UI updates. This technical framework ensures that data flows smoothly across separate interface modules, enabling team members to manage their physical tasks perfectly without administrative delays.

To expand on this technical architecture, Kumar et al. explicitly detail how multi-device state tracking safeguards small food businesses from accidental double-booking or missed time slots during extreme production rushes. The authors focus a significant portion of their research on the "state progression model" within food processing workflows, noting that custom baking requires multiple specialized phases—from raw material measuring to final icing decoration—which cannot be efficiently tracked using static page loads. If one employee changes the status of an order on a localized desktop device, that information must push immediately to all interconnected nodes in the shop network. By utilizing event-driven streaming data networks, systems create a live communication bridge that cuts through physical wall barriers inside an active workspace. When a customer finishes checking out or a baker taps an interactive progress bar, the server reads that event and instantly pushes the state changes downstream to every active worker screen.

Furthermore, the researchers delve into the technical management of high-volume data traffic during peak operating hours, a common pain point for small food enterprises. Kumar et al. argue that standard request-response network protocols put unnecessary stress on local network bandwidth because devices must constantly send "pings" to the server to check for new information. By moving to an event-driven setup powered by continuous document listeners, the data only moves when an actual change happens—such as a new order coming in or a payment status getting approved. The study shows that this method slashes network data consumption by more than half, allowing web applications to maintain a stable, lightning-fast connection even when running on affordable, consumer-grade internet plans in regional areas.

In addition to bandwidth efficiency, the study highlights how a live, multi-device tracking setup boosts accountability and reduces friction among employees. When every single update is timestamped and instantly pushed to all active screens, it creates a transparent digital trail of the entire production line. Kumar et al. track how this automated messaging layout cuts down on the need for staff to constantly double-check details with each other out loud, allowing front-counter cashiers and back-end kitchen staff to work independently yet in

perfect unison. The paper concludes that replacing paper notes with live, synchronized digital tracking screens is the most effective way for small food businesses to prevent order slip-ups, protect their daily profits, and maintain customer trust during heavy sales rushes.

This research study aligns perfectly with our Capstone project because it offers a concrete technical solution to solve C&N Bakery's disorganized, paper-based order system. Currently, when a customer sends a message on social media to buy a cake, the front counter employees have to write the instructions down on paper, walk over to the kitchen, and pin it to a corkboard. This clumsy, old-fashioned routine easily results in lost order slips, incorrect cake designs, and high-stress arguments among the kitchen staff when holiday rush seasons arrive. By applying the real-time tracking concepts proved in Kumar's study, our system connects the customer's checkout form straight to the manager's workspace using Firebase's live synchronization capabilities. The exact second a customer clicks submit online, a live notification card instantly pops up on the kitchen's desk monitor without requiring any manual web browser refreshes.

Additionally, this study serves as the direct foundational guide for our 4-stage kitchen stepper pipeline, which guides orders through the Preparing, Baking, Packing, and Delivery phases. As

the bakers complete their tasks and click through the digital steps on their workshop tablet, those data updates are blasted over Firebase to immediately change the status shown on both the customer's private account tracker and the front clerk's monitoring station. This creates a unified data loop across the whole bakery. By eliminating the communication gaps that happen when people try to pass verbal messages in a noisy kitchen, our system guarantees that all employee monitors show identical data at any given millisecond. This reduces workflow confusion, shortens waiting lines for order pickups, and gives the kitchen team a calm, highly organized workspace to handle heavy daily orders accurately.

Ramadhan and Sani (2025), in their research paper **"Empowering MSMEs with Data-Driven Insights: Mobile Sales Dashboard Application for MSMEs"** (ISSN): 2476-8812 , underscore that backend data aggregation code can automatically parse complex, multi-field transaction arrays to feed visual data visualization dashboards. This background computing process completely removes the tedious chore of forcing business managers to manually input numbers into spreadsheet programs, allowing them to instantly view dynamic graphs that map out sales trends and business growth metrics.

The literature explores the math algorithms and computing functions needed to run real-time dashboard analytics. Ramadhan and Sani track how raw, single-line database entries (like individual sales receipts) can be grouped, sorted, and computed inside the database using background aggregation pipelines. Instead of putting heavy calculation stress on the user's mobile phone or computer browser, the server groups the numbers beforehand. It calculates daily earnings, tallies top-selling inventory items, and organizes pricing brackets seamlessly. The authors conclude that presenting business data through clear visual components—such as vertical bar charts and horizontal flow graphs—allows non-technical business owners to easily spot underperforming assets, cut down on inventory waste, and make

smarter pricing decisions.

To expand heavily on these analytics capabilities, Ramadhan and Sani provide an extensive evaluation of historical data storage paradigms for small-scale commercial operations in provincial regions. The authors note that data accumulation without structuring becomes a technical liability for non-technical retail managers. When small businesses expand their menu lines to support customizable orders, tracking the raw numbers across multiple items breaks down if the data relies on manual bookkeeping or unlinked spreadsheet documents. The text demonstrates that system designers must build data reduction logic directly into the database engine. This prevents the user interface from choking on large history files and allows the platform to summarize months of transaction records in fractions of a second.

Furthermore, the researchers highlight the economic value of "predictive visibility dashboards" for micro-bakeries and pastry houses. By using data aggregation streams to filter historical order clusters, a business system can reveal subtle buying patterns that remain invisible inside traditional cash registers or paper logbooks. Ramadhan and Sani outline that these data streams can automatically group sales data by product type, design choice, and specific cost brackets. This automated

organization gives business owners the power to quickly recognize changes in customer preferences before ordering their next batch of baking ingredients. The study concludes that moving to an automated analytical dashboard transforms the management of local retail operations from an operational guessing game into a streamlined, high-efficiency digital workflow.

Additionally, the researchers emphasize the critical importance of mobile responsiveness and UI cleanliness in dashboard systems built for non-technical users. Ramadhan and Sani state that if an analytical workspace displays crowded datasets or requires complex filtering strings, business owners will quickly abandon the tool in favor of old manual routines. To combat this, system backends must automatically pre-calculate heavy mathematical values—such as net profits and weekly growth percentages—before passing the numbers to the frontend interface. By doing so, the application can populate simple summary blocks and responsive layout components instantly, ensuring that even users with zero background in data science can accurately interpret their business performance at a single glance.

Moreover, the study evaluates the security and integrity of automated financial tracking platforms. When financial calculations are computed automatically inside a secure database

script rather than being keyed in manually by local store clerks, the risks of internal fraud, missing receipts, and typos drop down to zero. Ramadhan and Sani document that automated tracking guarantees that every peso processed at checkout matches the database's sales charts perfectly. This background validation loop gives small enterprise managers total peace of mind, freeing them from the exhausting task of cross-referencing paper receipts at the end of every operating night and allowing them to focus entirely on expanding their product lines.

The text further addresses the long-term data scalability problems that small local operations face when their customer base grows over months of continuous business. Ramadhan and Sani point out that as sales records pile up, reading thousands of raw history lines can severely delay screen loading times if the application relies on older sorting models. The researchers suggest using modern document databases to slice data into pre-calculated chunks based on specific time intervals like daily, weekly, or monthly logs. This approach ensures that when an admin requests a long-term financial report, the platform pulls ready-made summaries instead of sorting raw rows on the fly, keeping the performance smooth and reliable.

This research study is highly relevant for our Capstone project because it provides the exact foundation we need to build our Revenue Management Canvas feature. Currently, the owner of C&N Bakery has to spend hours at the end of every week adding up paper receipts by hand to see if the store made or lost money. This manual bookkeeping routine is exhausting and often leads to financial math errors. Our platform removes this headache by using automated background math logic inside Firebase. Every single time a transaction goes through our website, Firebase automatically updates the numbers, turning messy sales logs into instant visual charts like vertical monthly revenue bars and horizontal top-selling product graphs. This setup makes it simple for the owner to track sales and see exactly which cake sizes, designs, or pricing brackets (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, and ~~P~~501+) bring in the highest profits without any accounting stress.

Furthermore, the study's discussion on inventory tracking perfectly fits our application goals. By looking at the automated dashboard charts on our platform, the owner of C&N Bakery can immediately spot which cake items are the most popular without searching through piles of old handwritten logbooks. This clear visual data helps them know exactly how many baking supplies, such as flour, sugar, and specialty icing

toppings, to order for the upcoming week. This automated tracking prevents the bakery from wasting money on dead stock that might spoil in the storage room. By changing the old, manual routine into a clear, visual workspace, the owner saves time, avoids management guesswork, and can plan ingredients smoothly while watching the business grow with complete accuracy.

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2.4 Synthesis of the Study

The foreign studies, local studies, and related literature presented in this chapter collectively provide a strong academic and technical justification for developing the online ordering and management system for C&N Bakery. Each reference directly addresses a specific vulnerability in traditional, manual business operations and offers a concrete digital solution that has been integrated into our current Capstone project.

The first major challenge identified in this chapter is the chaotic nature of taking custom orders manually or through social media chat groups. The local study by Ballena et al. (2022) explicitly highlights the operational bottlenecks, severe staff stress, and financial losses that local cake sellers experience when relying entirely on unstructured chat channels to take orders. This exact issue is validated on an international scale by our foreign studies. The first foreign study by Smith (2022) proves that manual ordering lacks data structure, which causes massive communication gaps between clients and bakers. The second foreign study by Davis (2023) adds that without an automated tracking tool, food enterprises constantly suffer from scheduling conflicts and overbooked calendar dates.

Our project directly solves these global and local problems by moving C&N Bakery away from messy text logs and introducing a structured web environment. To eliminate the communication gaps proven by Ballena et al. (2022) and Smith (2022) to cause product waste, our system implements an online storefront with dynamic ordering paths. By translating user choices directly into fixed data configurations based on the framework models established in the related literature by Tennakoon et al. (2022), the system removes human guesswork and locks in crucial design details before production begins. Furthermore, the timeline anxieties and overbooking risks detailed by Davis (2023) are solved by our Smart Scheduling Matrix Calendar, which mirrors the live tracking and queue-control methodologies studied by Salokhe et al. (2026) to manage the kitchen's daily capacity and stop timeline errors.

The second major area discussed by the researchers is market competitiveness and business insight. To ensure a pastry business remains competitive, the local study by Omowole and Sari (2025) emphasizes that modern Filipino consumers demand clear pricing layouts and active digital engagement. Our project directly applies this insight by introducing budget-friendly storefront filtering options divided into clear pricing tiers (P0-P199, P200-P500, and P501+). This removes purchasing friction

for the customer. Omowole and Sari (2025) also note that independent bakeries struggle because they lack data collection software to monitor their sales trends. This local struggle matches the findings of our third foreign study by Martinez (2024), which points out that bakeries globally lose significant profit margins because they do not track their best-selling items or financial metrics digitally.

We solve this exact problem by applying the data aggregation and sales summary principles discussed in the related literature by Azzahra et al. (2025). Our system automatically processes raw transaction inputs in the background, transforming cold data rows into a visual Revenue Management Canvas complete with monthly sales charts and top-product graphs, giving the owner instant business insights without manual arithmetic errors.

Finally, the technical execution of our platform relies heavily on the cloud frameworks and real-time networking principles established in the reviewed literature. The technical research by Tennakoon et al. (2022) and Salokhe et al. (2026) supports our choice of a cloud-native, serverless NoSQL database model. This ensures that C&N Bakery does not have to pay expensive monthly hardware upkeep costs, while still allowing the system to store highly complex, customized order details inside

flexible data structures that will not crash during high-traffic holiday rushes.

Once those orders are safely saved, the system keeps the entire business connected by using the live event-driven synchronization models detailed in the dashboard and canteen tracking applications from Salokhe et al. (2026) and Azzahra et al. (2025). The exact second a customer checks out online, a live notification pops up on the back-office monitor. The kitchen staff can then seamlessly track and move the order through our 4-stage kitchen stepper pipeline (Processing, Baking, Icing, Ready for Pickup).

In summary, by piecing all 3 foreign studies, 3 local studies, and 3 related literatures together, it is clear that transitioning from a manual, chat-based workflow to an integrated, real-time web application is necessary for C&N Bakery. Our Capstone project bridges the gap between traditional baking and modern software engineering, creating an automated ecosystem that protects profit margins, eliminates communication errors, and streamlines the bakery's daily workflow.

CHAPTER 3

3.1 Research Design

Our study employed a **descriptive-developmental research design**. This approach combines two important methods: studying a real problem in detail (descriptive) and building a practical software solution to fix it (developmental). This dual method works perfectly. It ensures that the web application we build is based on actual, real-world business problems and follows reliable software coding rules.

To give a clear and complete picture of how this study was conducted, we have detailed the exact definitions, practical uses, and methods we chose to leave out below.

What It Means:

Descriptive research is all about looking closely at a situation exactly as it is right now, without trying to change anything. Instead of testing theories or setting up artificial experiments, this method takes an accurate snapshot of daily operations. It maps out how tasks are done, documents the work habits of the staff, and points out the exact bottlenecks that slow things down. It answers the basic questions of what, who, where, and how a process works so that developers do not make

wrong guesses when writing code.

How We Used It in Our Project

We used descriptive research to study, understand, and document the manual struggles of C&N Bakery before writing a single line of code. We spent time at the bakery observing daily counter work, interviewing the business owner, and looking over their physical paperwork. This step gave us the exact details we needed to design our website, revealing these major pain points:

- **Messy Order Tracking:** Customer inquiries were scattered everywhere—across Facebook Messenger chats, SMS texts, and phone calls. Staff had to scroll through endless, unorganized chat histories just to find a single design idea, price quote, or delivery date.
- **Lost Order Details:** Complex cake details like flavor choices, icing types, sizes, and custom name inscriptions were easily miscopied onto loose scraps of paper or forgotten completely during busy kitchen shifts, leading to costly mistakes.
- **Calendar and Delivery Confusion:** Pickup dates and delivery schedules relied entirely on a handwritten notebook. Without an automated calendar to protect them, the bakery suffered from human scheduling mistakes, double-bookings,

and immense stress during major holidays.

- **Exhausting Financial Reviews:** Tracking down-payments, verifying GCash transfers, and adding up cash numbers were done manually with calculators and paper ledgers. This manual math routine took up too much time and increased the risk of accounting errors.

Why It Matters to Our Study

Without this descriptive phase, our web platform would have no real-world grounding. This step allowed us to turn general business complaints into exact software features. It helped us map out exactly how money changes hands and calculate the physical workload the kitchen staff can handle, ensuring our final application solves real business problems.

Development Research:

Developmental research is the step-by-step process of designing, building, coding, testing, and perfecting a software application over a set timeline. The main goal of this method is to take the raw notes and problems gathered from the descriptive phase and turn them into a practical, working tool. It is a continuous loop where early versions of the website are built, tested, reviewed by the actual users, and adjusted until the software runs smoothly and reliably under real-world conditions. This

approach prioritizes useful features, system speed, data safety, and clean screens, showing exactly how smart coding choices can solve everyday business headaches.

How We Used It in Our Project

Developmental research is the core technical muscle of this capstone project. It is the phase where we shift away from just talking about problems and actually start writing code to build a real solution. Using the exact lists of complaints and rules we gathered from our descriptive observations, our team designed and programmed the custom cake ordering platform, the smart booking system, and the sales dashboard.

This build phase involved creating two main website sections that connect to a single cloud database:

- **Building the Customer Portal:** We designed an online store with an easy-to-read menu split into clear categories: Cakes, Cupcakes, and Kakanin. To help budget-conscious buyers, we added simple price filters (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, and f~~P~~501+). We also coded the 4-choice Custom Cake Builder, which allows users to mix and match layers, flavors, frostings, and themes with an automatic price calculator. Customers can securely upload reference photos and look at a personal order status tracker (Pending, Unpaid, or

Cash-on-Delivery) to follow their orders.

- **Constructing the Owner's Dashboard:** We built a secure administrative workspace protected by a login screen so only authorized staff can get in. Inside, we assembled clear dashboard summary cards that give the owner a quick look at daily business activities. We also added a data visualization suite that turns raw database entries into easy-to-read vertical monthly sales charts and horizontal top product graphs.
- **Coding the Smart Business Logic:** We programmed background system rules to make both portals talk to each other safely. The primary rule is our smart scheduling calendar, which automatically counts incoming orders and instantly blocks out any date that hits a maximum limit of three (3) custom orders. We also integrated a 4-stage horizontal progress timeline to track a cake's journey through the kitchen using manual progress buttons (Prepare, Bake, Pack, Ship, and Cancel).

Why It Matters to Our Study

The developmental phase is what gives our project its value in the field of information technology. It serves as a real-world example of how affordable cloud database networks can be customized to support small, local neighborhood bakeries. By

documenting how smart coding rules can stop everyday human mistakes—like shutting down a calendar date when the kitchen is full or putting order details right next to customer reference images—this phase proves how custom-made software can protect and upgrade small retail businesses.

Why We Did Not Use Experimental Research

Experimental research is a strict lab method used to find cause-and-effect links. It works by changing one specific detail while keeping everything else exactly the same, then measuring the result using complex statistics and math tests. It usually requires dividing people into a "test group" and a "control group" to see how they react.

Why It Does Not Fit Our Project

This method does not work for our study for a few reasons:

- **No Variables to Change:** We are not running laboratory tests to see how changing oven heat levels or ingredient brands affects baking speeds. Our actual goal is to build a working web application that fixes real office problems.
- **Building Over Theory:** This is an IT capstone project focused on building a useful tool, not a science experiment. Success means making sure the website is safe, fast, and easy for C&N Bakery to use, not testing abstract

math formulas.

- **Bad for Daily Business:** Splitting real bakery customers into a "unlucky group" forced to use slow, messy paper notes and a "lucky group" using the new website just to measure their frustration would disrupt daily shop sales and annoy buyers.
- **Focus on System Integration:** Since the project is meant to replace an old, manual tracking style with an all-in-one cloud setup, the correct path must be developmental. We need to focus on coding secure database gates and price calculators, not running clinical lab controls.

3.2 System Development Methodology

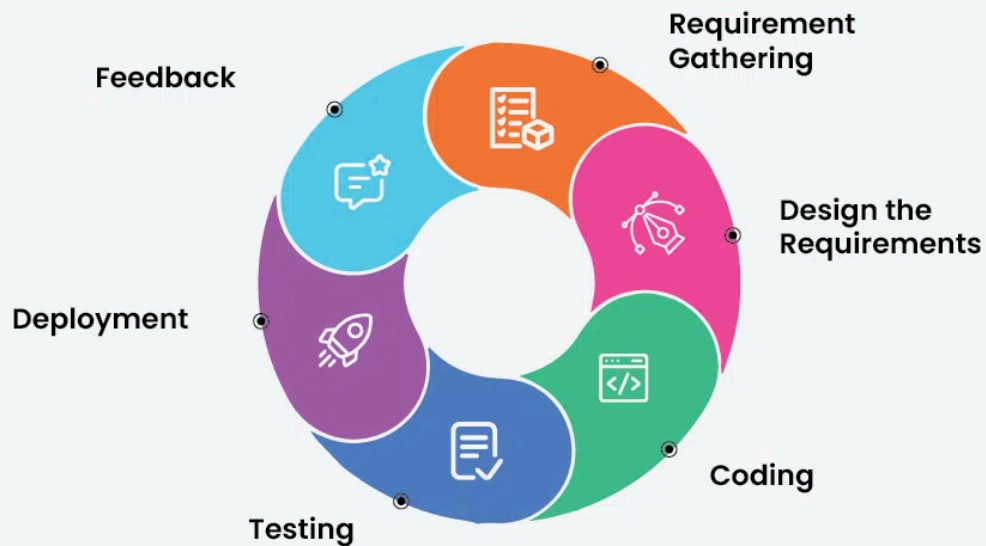
The researchers adopted the Agile Model as the primary Software Development Life Cycle (SDLC) framework for this project.

Because modern web development projects for small enterprises frequently encounter shifting requirements as users interact with early software builds, a rigid, linear development model can be highly inefficient. The Agile framework emphasizes adaptability, open communication, and the continuous delivery of working software modules through short, focused development cycles called sprints.

To ensure complete clarity regarding how this methodology was applied to the project, each phase of the framework is detailed below with its definition followed by an explanation of its execution within the context of the C&N Bakery platform.



Phases of Agile SDLC



1. The Planning Phase

- **General Definition**

The planning phase is the first step of the project where you figure out the core goals, decide what features the software will have, check what tools are available, and set up a clear calendar timeline for building the website.

- **How We Used It in Our Project**

During this phase, our research team sat down with the owner of C&N Bakery to establish exactly what the new website would handle and what tasks it would leave out. We mutually agreed that the web application should focus entirely on fixing the bakery's immediate problems: custom cake design choices, automated price calculations, calendar booking protections, and helpful sales graphs.

To keep the project manageable and avoid wasting time, we intentionally excluded unnecessary features like complicated delivery truck route mapping or automated wholesale flour and ingredient buying. During this window, we also mapped out our sprint calendar, ensuring that vital safety rules—like database security gates and the maximum order calendar locks—were scheduled to be built first so the bakery's kitchen workflow would be protected from the very start.

2. The Requirements Analysis Phase

- **General Definition:** This phase is all about figuring out exactly what the website needs to do and how it should behave. The development team gathers, organizes, and writes down a checklist of every single feature and performance rule required to fix the user's business challenges.
- How We Used It in Our Project

Our team broke down the bakery's daily operations into two clear checklists to make sure we did not miss any important features:

- **What the System Must Do (Functional Requirements):** The website must have a public store page split into clear sections for Cakes, Cupcakes, and Kakanin, using helpful price buttons (P0-P199, P200-P500, and P501+). It must include an interactive Custom Cake Builder that calculates prices instantly as choices change, a secure button to upload design photos, and a 4-stage kitchen tracker (Preparing, Baking, Packing, and Delivery). Finally, the admin calendar must have a strict limit that blocks out dates after three (3) custom orders are placed.
- **How the System Must Perform (Non-Functional Requirements):**
The website must be fully responsive, meaning it has to look clean and work perfectly on any smartphone, tablet, or computer. It must use a secure login wall to protect

sensitive financial records, update database numbers instantly without refreshing the page, and load fast so customers do not get frustrated and leave the site.

3. The Design Phase

General Definition

The design phase is where you turn your feature checklist into a clear technical blueprint. Before building anything, developers sketch out what the screens will look like and create a map showing how different pieces of information will connect and talk to each other in the database.

How We Used It in Our Project

Our team mapped out the complete structural blueprint of the platform before writing a single line of code. We drew user interface wireframes (rough screen layouts) to plan exactly where buttons, menus, and charts would sit on both the customer page and the owner's dashboard.

We also designed a clean database structure to keep information organized and linked correctly:

- **The Ordering Layout:** We mapped out how a customer's choices, custom text inscriptions, and uploaded cake photos

all link directly to one unique Order ID number so nothing gets mixed up.

- **The Scheduling Logic:** We structured the database to count active order rows for each calendar date. This gives the system the quick math it needs to automatically lock out full dates during checkout.
- **The Financial Setup:** We grouped transaction rows by payment status (Paid vs. Unpaid) and payment method (Cash on Delivery, GCash, or Bank Transfer) so the data streams feed perfectly into the dashboard sales charts.

4. The Implementation (Development) Phase

General Definition

The implementation phase is the actual building and coding step. This is where developers write the source code, connect the website to the database, and turn the design blueprints into a real, working application that users can click on and interact with.

How We Used It in Our Project

Our team programmed the entire web application using HTML5, CSS3, and JavaScript, connecting the front screens directly to a fast, serverless Firebase cloud database. We built everything in short, focused work windows:

- **First (The Customer Experience):** We focused entirely on coding the public storefront. We built a responsive menu catalog, dividing products into Cakes, Cupcakes, and Kakanin so customers could browse easily on their phones or computers. We then programmed the budget range-bound selectors (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, ~~P~~501+) and the 4-choice Custom Cake Builder. This included writing a live JavaScript pricing script that calculates costs in real time as users change cake layers, flavors, and frostings, along with a secure asset uploader for customer reference images.
- **Second (The Management Backend):** We dedicated our time to coding the private administrative workspace, protecting it behind a secure authentication login gate. We built the Smart Scheduling Matrix Calendar, programming the database rules to count active rows for each date and automatically lock out checkout dates once the bakery hits its maximum limit of three (3) custom orders. We also programmed the 4-stage horizontal progress pipeline (Preparing, Baking, Packing, and Delivery) so staff could move orders forward using manual action buttons (Prepare, Bake, Pack, Ship, and Cancel).
- **Third (Data Analytics & Alerts):** We focused on turning raw transaction records into business intelligence for the owner. We assembled the Dashboard Summary Cards for

high-level business tracking and integrated a Data Visualization Suite to render vertical Monthly Revenue Charts and horizontal Top Products Charts based on actual database entries. Finally, we set up automated background scripts to instantly change order flags (Pending, Unpaid, Cash-on-Delivery) and trigger email status updates to keep customers informed.

5. The Testing Phase

What It Means in Simple Terms

The testing phase is a strict quality check where developers run the software under different conditions to find hidden bugs, verify system rules, and make absolutely sure the platform works exactly how the users need it to before launch.

How We Used It in Our Project

Our team tested the website systematically across multiple layers during every single sprint to catch errors early:

- **Testing Single Features (Unit Testing):** We ran isolated tests on individual scripts to ensure their specific internal logic worked flawlessly.
 - *Example 1:* We repeatedly clicked through the 4-choice Custom Cake Builder to verify that the live JavaScript

pricing script added up costs accurately every single time a customer changed layers, flavors, or frosting options (ex. verifying that selecting a premium chocolate flavor correctly added the exact premium upcharge to the base price).

- *Example 2:* We tested the range-bound budget filters (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, and ~~P~~501+) to make sure that clicking a specific tier strictly displayed items matching those prices and hid everything else.
- *Example 3:* We tested the login script for the administrator portal by deliberately entering incorrect passwords and empty spaces to ensure the security gate remained securely closed and correctly showed an error message.

- **Testing Combined Features (Integration Testing):** We verified that different independent parts of the system communicated and shared data with each other correctly.

- *Example 1:* We tested the photo upload button to ensure that when a customer uploaded a custom cake reference photo, the image file successfully saved to Firebase cloud storage while its web link stayed permanently attached to the exact right text invoice row in the database.

- *Example 2:* We tested how the checkout page talked to the scheduling calendar. We verified that the moment an order was successfully submitted, the system sent a data signal to the database row counter, updating the total order count for that calendar date.
- *Example 3:* We tested the integration between the administrative workspace and the customer portal, confirming that when a staff member clicked the manual "Bake" or "Ship" buttons on the admin dashboard, the change instantly updated the order status flags (Pending, Unpaid, Cash-on-Delivery) visible on the customer's personal tracker.
- **Testing the Whole System (System Testing):** We ran the entire application end-to-end as a complete package to ensure flawless data flow from start to finish.
 - *Example 1:* We simulated a complete customer journey: a user logs in, configures a custom cake, uploads an image, chooses a date on the calendar, and submits the order. We then immediately logged into the admin side to verify that the order appeared on the master list, the reference photo rendered perfectly side-by-side with the cake specifications, and the automated

background script sent out the correct confirmation alert.

- *Example 2:* We ran a stress test on the scheduling system by submitting three (3) dummy custom orders on the exact same calendar date. We then attempted to place a fourth order for that same day to guarantee the Smart Scheduling Matrix properly counted the database rows and completely locked out that date, rendering it unclickable for the next customer.
- *Example 3:* We completed an entire financial transaction loop—moving an order from "Unpaid" to a completed state—to confirm that the Data Visualization Suite pulled the new database row automatically, instantly updating the vertical monthly revenue charts and the horizontal top products charts without any system delays, data leaks, or screen crashes.

6. The Deployment Phase

What It Means in Simple Terms

The deployment phase is the final step where the finished and tested software is launched live on the internet. The system is set up on secure cloud production servers and officially handed over to the business owner so they can start using it for real-world daily operations.

How We Used It in Our Project

Our team published the finalized web platform using cloud hosting services, making the online storefront fully live and accessible to the public through any standard mobile or desktop web browser.

To complete the launch and hand over the system smoothly, we executed the following steps:

- **Setting Up the Hardware:** We loaded and configured the private administrative dashboard directly onto the bakery's tablets and management devices, verifying that the secure login access gates blocked unauthorized users on site.
- **Cleaning the Database:** We wiped the backend database completely clean of all fake test accounts, dummy custom orders, and simulated sales transactions that we used during our testing phases.
- **Initializing Real Business Data:** We filled the fresh, live database with C&N Bakery's actual menu items, current pricing structures, real product categories (Cakes, Cupcakes, and Kakanin), and active operational guidelines.

This final transition successfully moved the shop away from its slow, risky paper notebooks and into a fully automated, real-time cloud operation.

7. The User Acceptance Testing (UAT) Phase

What It Means in Simple Terms

User Acceptance Testing (UAT) is the final evaluation step where the actual business owners, employees, and everyday customers try out the live system. This is done to make sure the website is easy to navigate, works perfectly in real life, and delivers real-world value before everyone gives it their final stamp of approval.

How We Used It in Our Project

Our team held a dedicated testing session right at the bakery, inviting the business owner, the kitchen bakers, the counter crew, and a select group of regular customers to put the live platform to the test:

- **From the Customer's Side:** Regular buyers verified that they could smoothly browse through the sweet treats, tap the budget range buttons, design their own desserts using the 4-choice builder, and watch their cake travel down the progress pipeline without getting confused.
- **From the Bakery's Side:** The owner and staff tested the backend system to ensure it fit their workflow. They verified that they could open and inspect uploaded down-payment receipts, use simple switches to mark

ingredients or products as "out of stock," and click through the kitchen timeline buttons as bakes moved into the oven.

Gathering this direct, hands-on feedback helped us ensure that the entire system felt completely natural and user-friendly for the people running the business every day.

8. The Maintenance Phase

What It Means in Simple Terms

The maintenance phase is the ongoing support step after the software goes live. This phase focuses on monitoring the application, fixing any unexpected errors that pop up over time, keeping loading speeds fast, and making minor tweaks to help the system adapt to the changing needs of the business.

How We Used It in Our Project

Once the custom platform officially took over daily operations at C&N Bakery, our research team stayed close to provide post-launch technical support. We kept an eye on system behavior through a few critical checks:

- **Monitoring System Health:** We continuously checked the database logs to ensure information saved cleanly without dropping connections.

- **Verifying Safety Rules:** We monitored the smart scheduling calendar to make sure it actively protected the kitchen by shutting down dates the absolute second a third custom order row hit the database.
- **Checking System Alerts:** We verified that automated emails consistently reached customer inboxes without getting delayed or flagged as spam.

3.3 System Architecture

The proposed system utilizes a Serverless Cloud Architecture powered by Google Firebase. Unlike traditional client-server

models that require a manual, custom back-end stack (such as a local server running PHP and a separate database framework), this architecture allows the front-end interface to communicate directly with cloud services using secure, built-in network channels.

This serverless setup removes the administrative stress, high hardware maintenance costs, and slow loading speeds often found in traditional server hosting, splitting the platform into two user-friendly environments:

- **Client-Side:** A highly responsive web interface built with HTML5, CSS3, and Vanilla JavaScript. It runs natively inside the user's web browser, handling the visual layouts, budget buttons (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, and ~~P~~501+), and immediate calculations for the custom cake builder.
- **Backend-as-a-Service (BaaS):** Google Firebase acts as our fully managed, invisible cloud engine. Instead of us building a server from scratch, Firebase manages the heavy-duty operational logic, user safety, and information file systems through three core tools:

1. Firebase Authentication

General Definition:

A secure, cloud-hosted login validation service that provides built-in encryption, user session management, and access rule checks to secure personal profiles.

What It Does in Our System

This module functions as the secure gatekeeper for the platform. It handles user registration, password logins, and active sessions, keeping personal details locked down. When someone tries to access the manager panel, look at financial charts, or toggle products as "out of stock," Firebase Authentication evaluates their account rank on the cloud side. This ensures that standard customer accounts are completely blocked from viewing or messing with private business files.

2. Firebase Realtime Database

General Definition:

A cloud-hosted, highly scalable NoSQL database engine that stores application information as an organized data tree and syncs updates instantly across all connected screens using live internet connection lines.

What It Does in Our System:

This service acts as the central digital storage vault for the entire bakery. It keeps an organized, real-time log of every menu item, product category (Cakes, Cupcakes, and Kakanin), buyer history file, custom cake configuration, and scheduling rule. Because it updates instantly, the exact millisecond a customer clicks "checkout" from home, the new record is saved to the cloud. This automatically updates the kitchen dashboard tablet on the fly, showing the new order specifications side-by-side with the 4-stage horizontal progress pipeline without forcing the bakery staff to refresh their screens. It also powers the Smart Scheduling Matrix by counting database order rows per date, giving the system the math it needs to automatically lock out full calendar dates.

3.4 Tools and Technologies Used

To construct, deploy, test, and manage the serverless platform, a specific stack of programming technologies, development tools, and database frameworks was chosen. The following sections detail each tool used.

1. HyperText Markup Language (HTML5)

General Definition:

The standard foundational language used to build and arrange content on the web. It uses semantic tags to tell the web browser where to place text headings, text boxes, buttons, media files, and data grids.

What It Does in Our System:

HTML5 acts as the structural skeleton of our entire application. It establishes the input forms inside the Custom Cake Builder, sets up the document frames for the split-screen order viewer, and creates the structured data tables for the administrator control panel. It ensures that raw database strings—like specific cake text inscriptions, customer phone numbers, and transactional histories—are presented clearly and accurately to both the bakery staff and public customers.

2. Cascading Style Sheets (CSS3)

General Definition:

A design styling language used to control the overall look of a website, including color schemes, typography, margins, layout padding, and how elements shift automatically to fit different device screens.

What It Does in Our System:

CSS3 controls the visual presentation and responsive behavior of our public storefront and private manager dashboard. It handles the mobile-friendly design, automatically turning a dense administrative desktop screen into a simplified, cleanly stacked vertical layout on a mobile tablet. It also styles the 4-stage kitchen tracking stepper timeline (Preparing, Baking, Packing, and Delivery), makes live pricing adjustments stand out, and applies the color-coded labels that instantly separate Paid receipts from pending Cash-on-Delivery collections.

3. Vanilla JavaScript

General Definition:

The core browser programming language used to build interactive features, adjust web document structures on the fly, run client-side math calculations, and exchange data over the network without relying on bulky external frameworks.

What It Does in Our System:

JavaScript serves as the logical engine running the frontend of our website. It drives the live calculator script, instantly changing the checkout quote as a customer taps different layers, flavors, or frostings. It also scans order records to count submissions per date to power the Smart Scheduling Matrix calendar locking gates, and runs the quick text-search feature on the admin table. Most importantly, it handles the direct API connections to Firebase, converting text inputs and image files into formatted JSON data streams without needing a traditional backend language like PHP.

4. Firebase Authentication

General Definition:

A secure, cloud-hosted identity validation service that provides built-in encryption, user session management, token handshakes, and access rule checks to secure personal profiles.

What It Does in Our System:

This service functions as our digital gatekeeper. It handles new user sign-ups, password changes, and active logins, keeping personal profiles safe from data breaches. When a user tries to access private sales charts, view financial records, or toggle a menu item's in-stock status, Firebase Authentication checks their account rank on the cloud side, ensuring standard customer profiles are completely blocked from viewing or changing private business records.

5. Firebase Realtime Database

General Definition:

A cloud-hosted, highly scalable NoSQL database engine that stores application information as an organized JSON tree structure and syncs changes instantly across all connected screens using live web socket communication lines.

What It Does in Our System:

This service acts as our central business storage vault. It archives every single menu option, product category (Cakes, Cupcakes, and Kakanin), buyer history file, custom cake configuration, and scheduling rule as a structured JSON object. Because it syncs in real time, the exact millisecond a customer clicks "checkout" from home, the record is saved to the cloud, immediately popping up as a brand-new order card on the kitchen's tablet dashboard without forcing anyone to manually refresh the screen.

6. Firebase Cloud Storage

General Definition:

A reliable, high-capacity cloud blob storage platform designed to let web applications upload, save, and retrieve large user-generated files—like photos and documents—directly via web connections.

What It Does in Our System:

This serves as the platform's media storage archive. When a customer uses the Custom Cake Builder to upload a design sketch, or attaches a GCash payment receipt screenshot during checkout, this storage service saves the file in an

encrypted cloud directory. It then returns a secure image web link that is tucked cleanly next to the text order details in the Realtime Database, allowing the bakers to open up and view those references clearly inside their split-screen workspace boxes.

7. Visual Studio Code (VS Code)

General Definition:

A powerful, lightweight source-code editor and Integrated Development Environment (IDE) featuring smart syntax highlighting, automatic text formatting, built-in error tracking, and integrated workspace controls.

What It Does in Our System:

VS Code was our main engineering workspace used to write and organize the platform's source files. Our team used its programming tools to draft our HTML layouts, style our CSS rules, and program our JavaScript logic. Its built-in code checking warned developers about missing variables or punctuation typos immediately, keeping our software code clean and easy to maintain throughout each development sprint.

8. Git and GitHub

General Definition:

A distributed version control tracking ecosystem paired with a cloud-hosted code repository service, used to log individual software updates, preserve safe history snapshots, and manage collaborative team coding projects.

What It Does in Our System:

Git was used to log every development update made to our project's codebase, serving as a reliable backup network. Every time a major system module was completed, a safe snapshot of the working code was sent to GitHub. This allowed our team to test out risky new code modifications without ruining previously working features, keeping a clean, historical timeline of our software development as we worked through our sprint schedule.

9. Firebase Console Dashboard

General Definition:

A centralized, web-based management control panel provided by Google that lets developers inspect database traffic, review user accounts, configure security boundaries, and monitor live system performance logs.

What It Does in Our System:

Our development team uses this dashboard to oversee our underlying cloud networks. It lets us write database data access rules, monitor cloud storage space usage, and review real-time server activity logs. If an execution issue or file upload error ever happens, the console displays detailed server logs that point out exactly where the code or network rule failed so we can fix it immediately.

3.5 Data Gathering Techniques

To design a system that fits the daily needs of the bakery perfectly, we used specific data gathering methods. These techniques allowed us to collect real-world, accurate details straight from the source.

Each technique is broken down below with its simple meaning and an expanded explanation of exactly how we applied it to capture the unique rules of our system:

1. Interviews

General Definition:

A direct conversation where a researcher asks structured or open questions to an individual or group to gather detailed personal insights, firsthand experiences, and specific requirements.

How We Used It in Our Project:

We conducted sit-down sessions with the C&N Bakery owner, counter staff, and kitchen bakers. We asked them about how they calculated costs, what frustrations they experienced with unconfirmed payments, and how many custom orders they could safely bake in a single day. These conversations gave us the exact specifications needed to build our system features.

- *Applying it to the System:* : The owner explained that a standard day maxes out at three custom bakes; this conversation became the exact logical rule we programmed into our backend row counter to trigger the Smart Scheduling Matrix lockout. The staff also detailed their frustration with manual payment tracking, which led us to design the secure file upload button so customers could attach GCash receipts during checkout.
- *Operational Details:* During these talks, the kitchen staff explained that they often faced issues when customers asked for last-minute design changes over text messages, which would easily get buried or forgotten. To solve this, we used their feedback to create the custom text inscription box in the frontend interface. Now, whatever wording the user inputs goes directly into the permanent backend database record, showing up right on the kitchen tablet so there is zero confusion while frosting the cake.

2. Observation

General Definition:

A method where researchers watch real-world activities as they happen in their natural setting to record actual

behaviors, timelines, and procedural habits without interfering.

How We Used It in Our Project:

Our team spent time inside the bakery during peak business hours to watch how orders were taken over the counter, how bakes were tracked as they moved through the kitchen, and how daily revenue was calculated. By watching this live, we spotted significant bottlenecks—such as staff losing track of an order's status or writing tallies down on loose scratchpads.

- *Applying it to the System:* Witnessing the staff yell across the kitchen to check on a cake's progress inspired us to build the 4-stage horizontal progress pipeline (Preparing, Baking, Packing, and Delivery) controlled by simple admin action buttons. We also noticed the owner spending hours compiling paper bills at night, which directly led us to integrate the data visualization charts that automatically map out monthly revenue and top-selling products using live database logs.

- *Operational Details:* We also stood by the counter and watched how regular walk-in customers interacted with the staff. Many customers on a tight budget would spend several minutes asking, "What can I get for ₱300?", forcing the clerk to manually filter options from memory. This observation directly inspired us to build the front-end budget tiers (₱0-₱199, ₱200-₱500, and ₱501+). Now, customers can instantly sort the menu themselves, saving time for both the buyer and the cashier.

3. Document Analysis

General Definition:

The systematic review and evaluation of a business's existing physical or digital records, forms, and logs to understand their baseline rules, formatting styles, and data structures.

How We Used It in Our Project:

We collected and studied C&N Bakery's physical paper logbooks, handwritten customer order slips, base pricing charts, and raw ingredient lists. Reviewing these physical records taught us the exact data formatting, mathematical

steps, and business rules that the bakery had used for years.

- *Applying it to the System:* We used their physical order forms as a direct structural blueprint to build the input fields for the 4-choice Custom Cake Builder. We analyzed their handwritten pricing rules to map out the exact mathematical logic for our live JavaScript pricing script, ensuring that base layers, premium flavor additions, and frostings calculated into a final total accurately. Finally, we studied their product lists to establish our permanent database JSON categories, splitting the data cleanly into Cakes, Cupcakes, and Kakanin.
- *Operational Details:* While digging through old, stained paper receipts, we discovered that the bakery struggled to track which orders were already fully settled, which were paid via partial down-payments, and which were Cash-on-Delivery (COD). To clean up this messy process, we took their exact payment rules and converted them into visual CSS3 color-coded status flags (Paid, Pending, Unpaid) inside the admin panel. This digital system updates automatically when a transaction completes, replacing their old habit of marking paper pages with colored highlighters.

3.6 Testing Procedures

To make sure the C&N Bakery website worked perfectly, stayed safe, and was ready for daily business, our team tested it inside and out. We wanted to be absolutely sure that everything—from the quick math that adds up cake prices to the live updates sent to the kitchen—ran smoothly without any glitches.

Each testing phase is broken down below with its simple definition and the exact way we executed it on our system:

1. Unit Testing

General Definition:

A testing method where developers isolate and check the smallest individual pieces of a software's code—such as a single script, button, or formula—to make sure that specific piece performs its job perfectly on its own.

How We Used It in Our Project:

We broke our system down into individual scripts and tested them independently before letting them interact with the rest of the website.

- *Applying it to the System:* We ran isolated tests on the live JavaScript pricing script inside the 4-choice

Custom Cake Builder. We inputted different combinations of cake layers, flavors, and frosting selections to guarantee that premium upcharges were calculated accurately every single time.

- *Operational Details:* We also ran separate unit tests on the front-end budget filters. We deliberately clicked each tier (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, and ~~P~~501+) to verify that the system successfully isolated the items matching those prices and instantly hid everything else. Finally, we tested the administrator login form by entering blank spaces and incorrect passwords to ensure the encryption gate stayed securely shut and displayed the correct error alert.

2. Integration Testing

General Definition:

A phase where multiple independently tested code modules are combined and tested as a group to verify that they communicate, exchange data, and work together seamlessly without causing system conflicts.

How We Used It in Our Project:

We focused on the real-time connections—the way the front-end screen that the customer sees talks and shares data with our background cloud services managed by Firebase.

- *Applying it to the System:* We tested the photo upload button to ensure that when a customer attached a custom cake design sketch or a GCash payment receipt, the image file successfully traveled to Firebase Cloud Storage, while its secure web link stayed permanently attached to the correct text order row in the Realtime Database.
- *Applying it to the System:* We also tested the connection between the customer's checkout actions and the automated scheduling system. We verified that the exact moment a user submitted a completed order, a data signal was successfully sent to the database row counter, instantly modifying the available slots for that calendar date. We also checked the integration between the admin panel and the customer's order tracker, ensuring that when staff toggled buttons on the kitchen dashboard, the status flags instantly changed across the web socket network.

3. System Testing

General Definition:

An end-to-end testing process where the completely integrated software application is evaluated as a whole package to ensure the entire data flow, security layout, and user experience function flawlessly from start to finish.

How We Used It in Our Project:

Our team simulated a series of complete, live transactions from the perspective of both a customer sitting at home and a staff member working in the shop.

- *Applying it to the System:* We executed a full business loop: we logged into a customer account, built a custom cake, uploaded a GCash screenshot, picked an open date, and submitted the order. We then immediately opened the shop's tablet dashboard to verify that the order card appeared instantly, the reference photo rendered perfectly side-by-side with the specifications, and the automated background script triggered the confirmation notice.
- *Operational Details:* We also ran a deliberate stress test on our calendar safeguards by submitting three

(3) dummy orders on the exact same date. We then attempted to place a fourth order for that same day to guarantee that the Smart Scheduling Matrix accurately locked out the date, turning it gray and making it completely unclickable for any future customer. Finally, we moved those orders entirely through the 4-stage horizontal progress pipeline (Preparing, Baking, Packing, and Delivery) to confirm that the transaction successfully updated the vertical monthly revenue charts and horizontal top-product charts without a single screen crash or data leak.

4. User Acceptance Testing (UAT)

General Definition:

The final evaluative phase where the actual target audience—the business owners, daily employees, and real consumers—interact with the live system to evaluate its ease of use, practicality, and real-world value before final approval.

How We Used It in Our Project:

We brought the live platform directly into C&N Bakery and ran a dedicated testing session with the business owner,

the kitchen bakers, the counter staff, and a selected group of regular local customers.

- *Applying it to the System:* Customers were handed smartphones and laptops and asked to independently browse the menu, use the budget filters, and design a custom cake. They verified that the interface felt natural and that they could track their order progress easily without needing to call the store.
- *Operational Details:* Simultaneously, the owner and bakers operated the administrative dashboard on store tablets. They verified that they could open and inspect uploaded payment slips clearly, use the quick switches to mark items as "out of stock" when raw ingredients ran low, and move active order cards smoothly across the kitchen progress timeline. Gathering this direct, hands-on feedback allowed us to make final layout tweaks, ensuring the system was fully optimized for their actual daily workflow before the official launch.

3.7 Evaluation Criteria

Each evaluation criterion is broken down below with its simple definition, an explanation of how it applies to our system, and concrete operational details from our project:

1. Functionality

General Definition:

Checks whether a software application completely and accurately performs every single feature, calculation, and task it was designed to do.

How It Applies to Our System:

We evaluated the system to confirm that both the public storefront and the private management panel executed their tasks exactly as planned.

- *Operational Details:* On the customer side, we verified that the catalog applied the budget filters (₱0-₱199, ₱200-₱500, and ₱501+) accurately, and that the Custom Cake Builder calculated live pricing upcharges correctly as users selected layers and frostings. We confirmed that automated emails triggered immediately for order confirmations, down-payment verifications,

and pickup or delivery status updates. On the admin side, we checked that the sales analytics charts mapped out monthly revenue, and that the interactive split-screen workspace successfully updated the production pipeline across its stages (Pending, Confirmed, Baking, Ready, Completed, and Cancelled) while syncing across the global view container. We also verified that the instant database text search bar, the quick-action In-Stock switches, and the receipt payment verification toggles altered data dynamically.

2. Usability

General Definition:

A measurement of how easy, intuitive, and stress-free a website is for real people to navigate, understand, and interact with without requiring special training.

How It Applies to Our System:

We evaluated how quickly and comfortably everyday buyers and busy bakery staff could complete their tasks on the platform using different devices.

- *Operational Details:* For customers, usability meant ensuring they could choose a cake flavor, type out a

custom text inscription, select a pickup date, and upload a GCash receipt screenshot all on a single, clean checkout screen. For the bakery staff, we evaluated how easily they could read the incoming order details. Thanks to the responsive layout design, dense data grids automatically adapted into clean, legible lists when loaded onto the kitchen's mobile tablets, allowing bakers to tap the pipeline update buttons or toggle ingredient stock settings with a single touch while working.

3. Reliability

General Definition:

A standard that measures how consistently a software system runs over time without crashing, losing information, or freezing, and how well it recovers if a user makes a mistake.

How It Applies to Our System:

We verified that our connection remained completely rock-solid and that data was safely preserved even during sudden internet drops or high traffic.

- *Operational Details:* We tested the system by submitting multiple orders simultaneously to ensure that Firebase handled the concurrent traffic smoothly. Because the system runs on a serverless architecture, it handles heavy data loads without slowing down. We also confirmed that if a user accidentally disconnected mid-transaction, their personal profile records and shopping details were safely preserved in the cloud. If a connection dropped momentarily, the web socket lines re-established themselves automatically without dropping active order cards from the kitchen's tablet display.

4. Efficiency

General Definition:

A benchmark evaluating the speed and performance of a system, focusing on how quickly it processes data, handles search queries, and generates updates to save time and reduce manual workloads.

How It Applies to Our System:

We measured how much faster the automated cloud system performed daily shop operations compared to the bakery's old, slow paper logbooks.

- *Operational Details:* We monitored the data syncing speed and confirmed that when a customer clicked "checkout" from home, the new order record appeared on the kitchen dashboard within milliseconds, eliminating the need to manually refresh the browser. We also verified that the administrative text search bar filtered through hundreds of customer rows instantly as the user typed. Additionally, the automated calendar row-counter instantly calculated daily bookings, locking out fully booked dates immediately and saving the owner from cross-referencing paper planners.

5. Security

General Definition:

The defensive measures, access controls, and data encryption practices used to safeguard sensitive user information, passwords, and payment files from unauthorized access or digital leaks.

How It Applies to Our System:

We evaluated the system's defensive layers to ensure that personal customer accounts, business revenue records, and payment files stayed fully protected.

- *Operational Details:* We verified that Firebase Authentication successfully acted as a digital gatekeeper, checking account ranks on the server side so that standard customer accounts were completely blocked from opening or altering private administrative files. We also tested the system's input validation rules to ensure that malicious code could not be typed into the text inscription boxes or search bars. Finally, we confirmed that uploaded GCash payment screenshots were stored in a secure cloud directory accessible only via verified administrator links, keeping financial transactions completely private.

